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Bachelor Thesis

Domain Gap Analysis between Automatically Generated OSM→CARLA Maps and a Manually Modeled Reference Map in Ingolstadt

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Declaration of Originality

I hereby declare that I have written this thesis independently and without unauthorised assistance. All sources and aids used are indicated as such. This thesis has not been submitted in the same or similar form to any other examination authority.

Ingolstadt, 30.04.2026

Michał Dembski

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Michał Dembski
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Abstract

In autonomous driving research, synthetic environments are widely used to train and evaluate perception systems. However, discrepancies between map-generation processes can introduce a persistent domain gap between different synthetic environments. This thesis investigates the domain gap between two CARLA environments representing the same region in Ingolstadt: (i) an automatically generated CARLA map derived from OpenStreetMap via an OSM→OpenDRIVE→CARLA pipeline, and (ii) a manually authored reference map.

The domain gap is analyzed primarily through its **structural** component (road geometry, topology, lane connectivity, and junction structure). To ensure experimental integrity, the OSM extraction bounds are fixed, the full toolchain is versioned, and pipeline determinism is evaluated using an external audit. The audit reveals non-determinism at the byte level while preserving the audited topological counts ($CV = 0.0$ for road and junction counts), motivating a bounded analysis of observed structural variability under fixed inputs rather than any claim of intentional domain randomization.

Structural comparison between the auto-generated and manually authored Ingolstadt maps yields a whole-network geometry RMSE of 54.84 m, a matched-subset RMSE of 24.47 m (73% of sampled road points within a 50 m criterion), and a Hausdorff distance of 1,663 m after CRS-normalized rigid alignment. The auto map is 4.88× longer in total road length and has 6.55× more junctions. The curvature distribution of the auto map is substantially wider than the manual reference (standard deviation 0.865 vs. 0.055 m^{-1}), indicating high-curvature outlier segments absent in the hand-authored map. A lane and junction connectivity analysis finds that the auto map carries no road-level predecessor/successor link declarations, relying entirely on junction-based connectivity, while all declared links in the manual map are fully resolvable.

CARLA-based quality assurance is interpreted conservatively. The final generated XODR can be parsed and loaded by CARLA, but load success is treated as a necessary but insufficient quality gate. A controlled planar ablation of the promoted DEM-backed artifact shows visual failure in all tested variants: the original map, a flat-elevation-only copy, and a flat-elevation-plus-lateral copy all retain detached slabs, malformed junction patches, and broken road-surface continuity. Flattening therefore does not remove the dominant defect class, indicating that the current visual failure is primarily horizontal/topological rather than a DEM-only elevation artifact. The structural comparison remains valid because it is performed at the OpenDRIVE level and does not depend on visually successful CARLA rendering.

Perception evidence is bounded rather than complete. A separate Town10HD baseline confirmed sensor-rig readiness (20 frames, all callbacks successful), but this is bounded evidence of pipeline mechanics rather than evidence that the generated Ingolstadt map is perception-ready. Direct generated-map perceptual gap quantification and generalization experiments therefore remain future work. An orthogonal semantic-object gap is observed in the authoritative contract-run artifact: the four-category object counter records zero barriers, buildings, poles, and trees in the auto map versus 1,538, 992, 3,427, and 4,232 respectively in the manual reference. A subsequent pipeline fix confirmed that building injection now produces the expected objects; that result is retained as supplementary pipeline evidence and does not alter the authoritative measurement. Contrastive encoder training on structural tile metrics produces a non-collapsed latent space (primary GNN: cross-cosine = -0.160; extended MLP encoder on $N = 45$ tiles: cross-cosine = 0.124, permutation test $p < 0.001$, $K = 1{,}000$), demonstrating statistically significant separation of auto-generated and manually authored map embeddings; no downstream accuracy or transfer-feasibility claim is made.

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Nomenclature

Δt Fixed simulation time step in synchronous mode

$\mathbf{K}_{\text{undist}}$ Undistorted camera intrinsic matrix (pinhole model), taken from `K_undistortion`

$\mathcal{G}$ Road or lane connectivity graph

IoU Intersection over Union used for tile matching / gating

RMSE Root mean squared error

${}^{c}\mathbf{T}_{v}$ Homogeneous transform from vehicle frame to camera frame (*vehicle*→*camera*)

${}^{v}\mathbf{T}_{l}$ Homogeneous transform from LiDAR frame to vehicle frame (*LiDAR*→*vehicle*)

d_H Hausdorff distance

D_{KL} Kullback–Leibler divergence

s OpenDRIVE longitudinal coordinate along the reference line

Glossary

domain gap A measurable divergence between the training data distribution and the deployment data distribution, which can degrade model performance under distribution shift.

domain randomization A strategy for improving sim-to-real transfer by increasing variability in synthetic data (e.g., textures, lighting, assets, or scene structure) so that real-world conditions fall within the training distribution.

geoReference An OpenDRIVE header field that encodes the map's coordinate reference system (e.g., PROJ string) to enable consistent alignment with geographic coordinates.

laneLink OpenDRIVE predecessor/successor relationship used to describe how lanes connect across road sections and junctions.

laneSection OpenDRIVE element defining the lane cross-section of a road over a longitudinal interval, including lane identifiers, widths, and lane types.

planView OpenDRIVE element describing the two-dimensional reference-line geometry of a road, composed of geometric primitives such as lines, arcs, and spirals.

road-level link OpenDRIVE predecessor/successor declaration at road level, used to encode connectivity between roads independently of lane-level links.

Acronyms

API Application Programming Interface.

AV Autonomous Vehicle.

BEV Bird's-Eye View.

CARLA Car Learning to Act.

CRS Coordinate Reference System.

CSV Comma-Separated Values.

DEM Digital Elevation Model.

DNN Deep Neural Network.

GNSS Global Navigation Satellite System.

GPU Graphics Processing Unit.

HPC High-Performance Computing.

IoU Intersection over Union.

JSON JavaScript Object Notation.

KITTI Karlsruhe Institute of Technology and Toyota Technological Institute.

LiDAR Light Detection and Ranging.

mAP mean Average Precision.

mIoU mean Intersection over Union.

ML Machine Learning.

ODbL Open Database License.

OpenDRIVE OpenDRIVE (ASAM road network format).

OpenSCENARIO OpenSCENARIO (ASAM scenario description standard).

OSM OpenStreetMap.

QA Quality Assurance.

RMSE Root Mean Squared Error.

ROS Robot Operating System.

SE(3) Special Euclidean group in 3D.

SLURM Simple Linux Utility for Resource Management.

SUMO Simulation of Urban MObility.

UDA Unsupervised Domain Adaptation.

UTM Universal Transverse Mercator.

WGS84 World Geodetic System 1984.

XODR OpenDRIVE file format (commonly `.xodr`).

YOLO You Only Look Once.

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1. Introduction

Synthetic environments are widely used in autonomous driving research because they enable scalable, safe, and controllable experimentation. Yet perception models trained in simulation often struggle to generalize beyond their training conditions due to persistent domain gap effects Bai et al. (2024). While this problem is often framed as a simulation-to-reality issue, relevant domain differences can also emerge *within simulation itself* when the same geographic region is represented through different map-generation processes.

This thesis studies that intra-simulation gap for two large-scale CARLA environments of the same Ingolstadt region: an automatically generated map derived from OpenStreetMap (OSM) via an OSM–OpenDRIVE–CARLA pipeline, and a manually authored reference map. The primary emphasis is on reproducible and failure-aware quantification of structural differences in topology, geometry, and roadside semantics, while perception evidence is treated conservatively under a fixed sensor-rig contract.

Perceptual gap quantification (RQ3) is bounded rather than complete: the strict-evidence capture attempt on the auto-generated map targeted 50 frames but produced zero recorded frames due to a confirmed CARLA runtime boundary (streaming transport refused connections on the custom world). The sensor rig was separately confirmed correct on a built-in Town10HD map (20 frames, all 14 sensor callbacks successful), so the failed target-map attempt is interpreted as a CARLA runtime/resource boundary on the specific Ingolstadt setup rather than as evidence of a rig implementation error. Direct paired perceptual quantification remains future work.

GNN contrastive training (RQ4) was affected by representation collapse under a positive-only cosine loss; after switching to an NT-Xent objective the latent space was confirmed non-collapsed (cross-cosine $= -0.160$, verdict `NOT_COLLAPSED`). An extended contrastive encoder trained on $N = 45$ structural-feature tiles (30 auto / 15 manual) and validated by a $K = 1{,}000$ label-permutation test yields $p < 0.001$, confirming that the latent-space separation between auto-generated and manually authored tiles exceeds chance; this upgrades RQ4 from observational to statistically supported. No downstream transfer experiment was executed. Transfer feasibility (RQ5) is deferred accordingly.

The resulting claim scope covers: determinism characterization (RQ1), structural gap quantification (RQ2), bounded perception-infrastructure evidence (RQ3), statistically supported GNN latent-space separation (RQ4, permutation $p < 0.001$), and an explicitly deferred transfer question (RQ5). This thesis primarily resolves the structural OpenDRIVE-level domain-gap comparison; direct perceptual domain-gap quantification and transfer/generalization experiments remain deferred, while the GNN result is retained as a statistically supported structural diagnostic without any downstream accuracy or transfer-feasibility claim.

1.1. Relevance and aim

Map quality is not a cosmetic detail: it affects traffic flow, lane-level routing, the placement and visibility of infrastructure, and the distribution of objects encountered along a route. Therefore, differences between OSM-derived maps and manually authored maps can influence both the *structural* properties of a scenario and the *perceptual* statistics observed by sensors. The aim of this thesis is to (1) quantify structural domain gaps between OSM-generated and manual maps, and (2) validate perception-capture capability under strictly controlled experimental conditions, while deferring direct manual-vs-generated perceptual quantification until stable paired capture is available.

1.2. Research questions and working hypotheses

We address the following research questions:

- **RQ1 (Determinism):** Does the OSM→OpenDRIVE pipeline preserve stable structural and topological signatures under fixed inputs, and where does byte-level nondeterminism arise?
- **RQ2 (Structural domain gap):** How do automatically generated OSM-based CARLA maps differ structurally from a manually modeled CARLA map of the same region?
- **RQ3 (Perceptual domain gap):** How do these structural differences shift perception outputs when running an identical sensor rig and route protocol?
- **RQ4 (Structural variability and latent representation):** Does repeated automatic map generation from the same pinned inputs introduce measurable structural variability, and can a latent representation of that variability support robustness analysis?
- **RQ5 (Generalization and transfer):** To what extent do perception models trained on automatically generated maps generalize to (a) a manual simulated map and (b) unlabeled real-world data?

Our working hypotheses are that (i) geometry- and topology-level determinism is achievable with fixed seeds and pinned tool versions, even if byte-level artifacts differ due to serialization effects; and (ii) the largest structural gaps appear in semantics and lane topology rather than coarse road placement. A third hypothesis — that perceptual shifts correlate with structural errors such as misaligned lanes, incorrect junction connectivity, and missing traffic-control elements — was formulated but could not be evaluated within the thesis scope because direct paired perception capture on the auto-generated map was blocked by a CARLA runtime boundary (see the Scope Limitation paragraph below). A downstream transfer question (RQ5) was scoped conditionally on successful paired perception capture and is therefore reported as deferred rather than answered.

1.3. Research questions and falsification criteria

To ensure that claims in this thesis are testable rather than narrative, each research question is paired with the evidence used to answer it and explicit falsification criteria.

RQ1 (Pipeline determinism). Is the OpenStreetMap→OpenDRIVE→CARLA pipeline deterministic under fixed inputs, and if not, at which stage does nondeterminism arise?

- **Evidence:** repeated runs with pinned OSM extracts (SHA-256), multi-level hashes (byte-level and normalized geometry-level), and semantic/topological signatures (e.g., road/junction counts and lane distributions).
- **Falsification:** identical pinned inputs yield divergent semantic road topology or inconsistent CARLA-exported OpenDRIVE (`world.get_map().to_opendrive()`) without any change in software version or settings.

RQ2 (Structural domain gap). What structural differences exist between automatically generated Ingolstadt maps and a manually modeled large-area CARLA map of the same region?

- **Evidence:** comparative OpenDRIVE-derived metrics for topology (connectivity, junction density), geometry (curvature, lane widths), semantics (traffic control proxies), and tile seam/adjacency checks.
- **Falsification:** no statistically meaningful differences are observed between manual and generated maps under the chosen structural metrics.

RQ3 (Perceptual domain gap). How do structural map differences translate into perceptual differences for camera and LiDAR sensors?

- **Evidence:** sensor outputs recorded under a single authoritative rig (Section 5.14) using identical routes, weather, synchronous stepping, and tick budgets across domains; perceptual proxy metrics quantify distribution shift.
- **Falsification:** perceptual proxy metrics remain unchanged despite measurable structural differences, under identical sensor calibration and protocol.

This criterion could not be evaluated within the thesis period because direct paired capture on both maps was blocked by a CARLA runtime boundary (Section 6); RQ3 is therefore classified as BOUNDED pending execution of the paired experiment.

RQ4 (Structural variability and latent representation). Does repeated automatic map generation from the same pinned inputs introduce measurable structural variability, and can a contrastive latent-space model trained on that variability serve as an observational basis for robustness analysis?

- **Evidence:** structural metric distributions across repeated generations; GNN contrastive training (NT-Xent) on the resulting structural embeddings as an observational probe for representation quality; collapse detection via cross-domain cosine similarity.
- **Falsification:** repeated generations yield identical semantic signatures (no meaningful variability), or the contrastive latent space collapses (cross-cosine ≈ 1.0) indicating no domain-discriminative structure was learned.

Identical semantic signatures are operationally defined as $CV < 0.01$ for road and junction counts across repeated runs; values above this threshold constitute meaningful structural variability.

RQ5 (Generalization). To what extent do perception models trained on automatically generated maps generalize to (a) a manual simulated map and (b) unlabeled real-world data?

- **Evidence:** not collected within the thesis scope. The experiment infrastructure (data collection entrypoints, proxy metric definitions) is implemented but the transfer experiment was not executed; RQ5 is deferred to future work as stated in the scope limitation below.
- **Falsification:** no measurable shift is observed between evaluation domains, or observed shifts can be explained by calibration or protocol changes rather than map-domain effects.

Scope Limitation: Perceptual Domain Gap and Generalization. The original thesis brief assigned by the supervising institution required, in addition to structural gap analysis: (i) quantification of perceptual domain gap between generated and manual maps, (ii) evaluation of perception-model generalization on the simulated Ingolstadt area, and (iii) evaluation of generalization to unlabeled real-world data when training on automatically generated maps. These deliverables correspond to RQ3 and RQ5.

The present thesis delivers the structural foundation (RQ1, RQ2) and provides bounded evidence for RQ3 and RQ4, but the direct perceptual quantification and transfer experiments could not be completed within the thesis period for two specific technical reasons. First, the final promoted XODR—while parseable and loadable in CARLA—fails visual road-mesh QA: the controlled planar ablation shows detached slabs, malformed junction patches, and broken road-surface continuity even after elevation flattening, confirming that the dominant failure class is horizontal/topological. A generated map that fails visual QA cannot serve as the basis for a paired perception experiment. Second, the manually authored Grid0828 cooked-world capture path was bounded by a CARLA runtime boundary during live sensor spawn, preventing direct manual-versus-generated paired recording. These are engineering pre-condition failures, not design decisions. Consequently, Town10HD is used only as bounded evidence that the sensor rig and capture stack function correctly on a stable built-in map, while direct manual-versus-generated perceptual quantification (RQ3) and learning-based generalization experiments (RQ5) remain deferred to future work once the generated map passes visual QA (see Section 9.3).

1.4. Contributions

This thesis contributes:

- A reproducible experimental comparison between an OSM-generated map set and a manually modeled reference map of the same area in CARLA 0.9.16.
- A determinism protocol with multi-level hashing (byte-level and geometry/topology-level) and repeatability reports.
- Structural domain-gap metrics computed from OpenDRIVE (topology, geometry, semantics) and sensor-rig capability proofs under a unified, mathematically verified calibration contract (consistent extrinsic conventions for camera and LiDAR placement, verified against the CARLA coordinate frame).
- A bounded analysis of observed conversion variability across multiple OSM-derived map variants (reported as variability evidence, not as a performance-gain claim).

1.5. Thesis structure

Chapter 2 introduces background and related work on simulation, domain gap, and map standards. Chapter 3 presents foundational concepts: the OSM data model and its limitations, the Ingolstadt study area, the manual reference map family, sensor calibration conventions, and CARLA constraints. Chapter 4 describes the automated OSM-to-CARLA map generation pipeline: stage architecture (the implemented pipeline stages), quality gates, coordinate reference handling, artifact contracts, and the determinism and provenance strategy. Chapter 5 details the map generation and repair method: tile correspondence, operational drivability definition, topology repair, seam continuity, and sensor rig correctness. Chapter 6 covers simulator-based quality assurance and perception infrastructure validation, including the Town10HD rig confirmation and bounded perception evidence. Chapter 7 presents quantitative results: structural metrics (RMSE, Hausdorff, curvature divergence) defined and measured, RQ1 determinism audit, RQ2 structural gap, and bounded evidence for RQ3 and RQ4. Chapter 8 discusses implications for domain gap and generalization, connectivity fidelity, semantic object absence, and threats to validity. Chapter 9 presents conclusions and future work.

2. Related Work

We frame cross-map transfer primarily as a domain generalization problem, where the target domain is not used for supervised training; for a broader perspective, see the survey by Zhou et al. (2021).

This chapter situates the thesis at the intersection of (i) driving simulation and synthetic data, (ii) domain shift and sim-to-real generalization, and (iii) map representations and automated map generation.

2.1. Simulation platforms and synthetic driving data

CARLA is an open-source simulator designed for autonomous driving research, providing programmable sensors, controllable environments, and digital assets for urban driving tasks (Dosovitskiy et al., 2017). The availability of simulation platforms has enabled large-scale training data generation and controlled evaluation, but it also shifts the bottleneck toward *map availability* and *map stability*. City-scale maps are particularly challenging because mesh generation, navigation preprocessing, and routing assumptions can fail even when the input OpenDRIVE is structurally valid (CARLA Team, 2026).

2.2. Domain shift, sim-to-real, and adaptation

Domain shift describes the mismatch between training and deployment distributions. In autonomous driving, the domain gap can originate from appearance (textures, lighting), sensor physics (noise models), and structure (geometry/topology). Domain adaptation methods attempt to reduce this gap, for example by learning domain-invariant features via adversarial training (Ganin et al., 2016) or by aligning feature statistics (Sun and Saenko, 2016). Domain randomization instead injects variability into the simulator so that real-world inputs become "just another variation" (Tobin et al., 2017). These ideas motivate the thesis focus on whether repeated, automatically generated maps exhibit *natural* variability, and whether such variability can support more robust generalization.

Unsupervised image translation and cycle-consistency. Beyond feature-level alignment, image-level translation can reduce perceptual gaps by mapping simulated imagery toward the real domain. SimGAN refines synthetic images using adversarial training without requiring paired real images Shrivastava et al. (2017). CyCADA extends this idea with cycle-consistency and semantic constraints to preserve task-relevant structure during translation Hoffman et al. (2018). Even if such methods are not used for training in this thesis, they provide useful baselines for interpreting whether observed errors are driven primarily by appearance or by map structure.

Measuring distribution shift without labels. For evaluation on unlabeled real data, two-sample tests and embedding-space distances can quantify shift directly. The Maximum Mean Discrepancy (MMD) test is a standard kernel-based approach for measuring whether two sets of samples come from different distributions (Gretton et al., 2012). This motivates the thesis use of feature-space proxy metrics when real-world ground truth is unavailable.

Formal definition of the domain gap. In statistical terms, a *domain* corresponds to a data distribution over inputs and (optionally) labels. A *domain gap* (also discussed as *dataset shift* or *domain shift*) exists when the training distribution and the deployment distribution differ, i.e.,

$$P_{\text{train}}(x, y) \neq P_{\text{test}}(x, y). \tag{1}$$

Such shifts can be caused by differences in appearance (lighting, textures, weather), sensors (noise, intrinsics/extrinsics, resolution), or scene structure and semantics. (Quiñonero-Candela et al., 2009)

Sim-to-real context. In simulation-based autonomous driving, it is useful to distinguish (i) **perceptual** gaps (rendering and sensor models) and (ii) **structural** gaps (road geometry, topology, and object layout). A common mitigation strategy is *domain randomization*, where the simulator deliberately varies rendering and scene parameters to encourage robust feature learning (Tobin et al., 2017). This thesis does not apply intentional parameter randomization; instead it measures whether bounded structural variability arises under repeated OSM-based map generation, and whether that variability contains domain-discriminative structure (discussed in Chapter 7).

Contrastive representation learning as an analysis probe. Contrastive learning provides a self-supervised method for encoding structural differences into a compact latent space. SimCLR (Chen et al., 2020) shows that representations learned by maximizing agreement between augmented views of the same instance while pushing apart different instances are competitive with supervised baselines; the NT-Xent loss used there has proven robust against representation collapse (cross-cosine ≈ 1.0) across diverse domains. In this thesis, contrastive training (NT-Xent) is applied to structural metric distributions as an *observational probe*: the goal is not to train a production model, but to verify that embeddings of auto-generated and manually authored maps occupy distinguishable regions of latent space. A non-collapsed result confirms that the chosen structural metrics contain domain-discriminative information, providing a weaker but principled form of evidence for RQ4.

2.3. Map representations for driving simulation: OpenDRIVE

ASAM OpenDRIVE is a standard for representing static road networks for driving simulation using an XML schema with explicit geometry, lane sections, and connectivity ASAM e.V. (2026). While OpenDRIVE is well suited as an interchange format, downstream stacks can be fragile: small inconsistencies in lane connectivity or junction definitions can violate routing assumptions and lead to simulator failures CARLA Team (2026).

2.4. OpenStreetMap as a data source and quality considerations

OpenStreetMap is an open, volunteer-generated dataset (VGI) that encodes roads as polylines with tags and relations. Its strength is global coverage and frequent updates, but completeness and tagging consistency can vary across regions Haklay (2010). Early work on VGI stressed that errors and heterogeneity are inherent to crowdsourced geographic data Goodchild (2007), and subsequent quality-assurance literature proposes three complementary strategies — crowdsourcing, social, and geographic approaches — to identify and correct errors Goodchild and Li (2012). Comprehensive surveys of VGI quality assessment methods further emphasize that contributions originate from heterogeneous contributors using varied tools and that data may differ in positional, semantic, and thematic accuracy Senaratne et al. (2017). These findings motivate explicit uncertainty handling and conservative defaults in downstream conversion pipelines.

Previous studies quantify OSM completeness and positional accuracy and report strong region-dependent variability, which directly affects downstream conversion quality Haklay and Weber (2008); Girres and Touya (2010); Barron et al. (2014); Goodchild and Hunter (1997). Quality surveys note that VGI contributions can be geographically uneven, with heterogeneity in the level of detail, precision, and update frequency Senaratne et al. (2017), reinforcing the need for spatially aware quality checks and regional baselines.

2.5. Automated map generation and validation pipelines

Real2Sim presents an automated pipeline that generates simulator towns from OpenStreetMap and motivates this direction as a scalable benchmark generator for autonomous driving. It evaluates similarity between generated environments and real-world datasets using geometric and LiDAR-based footprint statistics, providing a useful reference point for the map-generation and evaluation framing in this thesis Mondal et al. (2020).

OSM-derived pipelines typically include: extraction of a bounded region, conversion into a simulator representation, topology repair, and validation. Some approaches rely on graph-based repair and enrichment, while others use external tools such as SUMO Behrisch et al. (2011a); Krajzewicz et al. (2012) netconvert for consistency checks. In CARLA, OpenDRIVE Standalone Mode enables loading a user-provided OpenDRIVE map via `client.generate_opendrive_world()`, but this workflow can be unstable for malformed topologies or very large maps CARLA Team (2026). The thesis positions itself as a *research-grade* pipeline: repairs are explicit and logged, invariants are enforced continuously, and evaluation is separated from simulator execution.

2.6. Reproducibility and deterministic scientific pipelines

Reproducibility is widely recognized as a minimum standard for computational research claims Peng (2011). Practical guidelines emphasize provenance logging, stable configurations, and artifact management Sandve et al. (2013). These principles matter directly for map generation: without determinism, domain-gap measurements can confound genuine differences with accidental pipeline drift.

2.7. Research gap

The gap addressed is that city-scale map generation requires more than schema validity: it needs explicit invariants aligned with simulator constraints, robust QA workflows resilient to nondeterministic crashes, and offline comparison metrics that remain usable when simulator validation is unreliable.

3. Foundations and Problem Setting

This chapter introduces the source and target representations used in this thesis and derives the constraints that define the problem.

3.1. OpenStreetMap as a source of road geometry and semantics

OpenStreetMap encodes roads primarily as `way` objects: ordered lists of `node` coordinates with tags describing road class and attributes. As a VGI dataset, OSM exhibits heterogeneous completeness and regional conventions Goodchild (2007); Haklay (2010).

In addition to the peer-reviewed VGI quality literature, this thesis follows the normative OpenStreetMap community definitions for the underlying data model and tag semantics. The OSM data model is based on *nodes*, *ways*, and *relations* with key–value tags OpenStreetMap Wiki contributors (2025a, 2026). One-way traffic is interpreted according to `oneway=*`, where `oneway=yes` follows the way direction and `oneway=-1` reverses it OpenStreetMap contributors (2024d). Turn restrictions are represented by `restriction` relations with `from/via/to` roles and a `restriction=*` value OpenStreetMap contributors (2024e). Where lane-related tags are available, lane counts and manoeuvre hints are derived from `lanes=*` and `turn:lanes=*` OpenStreetMap Wiki contributors (2025b); OpenStreetMap contributors (2024g). Grade-separation cues (`layer=*`, `bridge=*`, `tunnel=*`) are used to avoid illegal merges between roads at different vertical levels OpenStreetMap contributors (2024c,a,f). Roundabout semantics follow `junction=roundabout` guidance OpenStreetMap contributors (2024b). Licensing and attribution for all OSM-derived data follow the Open Database License requirements as stated on the official OpenStreetMap copyright and license page OpenStreetMap Foundation (2026).

For road-network conversion, three limitations are especially relevant.

- **Incomplete semantics:** lane counts, lane widths, lane types, and elevation are often missing or inconsistent.
- **Fragmented topology:** intersections can be represented by near-miss endpoints, missing connections, or node mismatches.
- **Tagging inconsistencies:** conventions vary across regions; even basic tags such as `oneway` and `lanes` can be absent or ambiguous.

3.2. Study area: Ingolstadt extraction bounds

The experiments focus on a fixed geographic region in Ingolstadt. The OSM extract is cropped using a deterministic bounding box, stored locally, and reused across runs to avoid variability from live downloads.

Table 1: Geographic bounds used to extract the Ingolstadt study area from OpenStreetMap.

Parameter	Value
Lat-Min	48.74935649548228
Lon-Min	11.422268084715878
Lat-Max	48.77444431571603
Lon-Max	11.47882091528412

3.3. Reference maps: manual baseline vs. automatically generated maps

This thesis compares two map sources representing the same geographic region:
(i) a manually created reference map of Ingolstadt ("manual map"), and
(ii) a family of automatically generated maps derived from OpenStreetMap ("auto maps").

The OSM-derived candidate map is generated as an OpenDRIVE artifact and evaluated through the CARLA workflow. The Grid0821/Grid0828 reference maps are manually modeled CARLA maps; their available OpenDRIVE/XML data are parsed offline for structural comparison. The manual map serves as a higher-fidelity baseline, as a curated CARLA reference, not as a complete ground-truth representation of all OSM road classes: it is produced through manual modeling and curation and is therefore expected to exhibit fewer structural and semantic inconsistencies than directly converted OSM data. The auto maps are produced by the proposed OSM→OpenDRIVE→CARLA pipeline, including topology repair and semantic completion steps designed to satisfy CARLA's drivability constraints.

To enable meaningful comparison, both sources are evaluated under a shared georeferencing and sensor setup (Sections 4.3 and 3.7). In particular, the study area bounding box (Table 1) is treated as the authoritative spatial scope. Any subsetting, tiling, or quality-gate evaluation performed later in the pipeline inherits this scope and is logged together with configuration snapshots and hashes for reproducibility.

Table 2 summarises the key structural properties of the three CARLA environments used in this thesis. Auto-generated geometry and road-length values are anchored to the authoritative run_11 governed artifact; junction-count and connectivity summaries come from the governed connectivity companion artifact family, and corrected curvature spread comes from the corrected support measurement on the same authoritative manual-vs-contract-run pair. Manual map values are computed from the OpenDRIVE XML files (`manual_grid0821.xodr`, `manual_grid0828.xodr`) and confirmed by CARLA stage-probe artifacts. Grid0821 and Grid0828 are both manually authored maps of the same Ingolstadt area and constitute the manual reference family; Grid0828 is the primary structural baseline used for the RQ2 domain-gap analysis.

Table 2: Structural overview of the three CARLA environments used in this thesis. Road counts and lengths for the manual maps are computed from the OpenDRIVE XML; spawn points are confirmed by CARLA stage-probe artifacts. Auto-generated road count and total road length are anchored to the authoritative run_11 artifact, while the auto junction count comes from the governed connectivity companion artifact family and the auto curvature standard deviation comes from the corrected support measurement on the authoritative manual-vs-contract-run pair.

Property	Auto-gen. (OSM)	Manual Grid0821	Manual Grid0
Data source	OSM, this pipeline	Manual authoring	Manual author
CARLA load mode	OpenDRIVE standalone	Cooked Unreal map	Cooked Unreal m
Road count	5,837	993	
Junction count	779	119	
Total road length	258.6 km	53.5 km	52.9
CARLA spawn points	run_11: n/a; load probes: 8,267–8,535	11,634	11,
Mean lane-section entries / road	1.14	43.9 (Grid0828)	
Curvature std (m^{-1})	0.865	0.055 (Grid0828)	
RQ2 structural baseline	Yes (run_11)	—	Yes (prima
Elevation (run_11 authority)	planar†	planar	pla

† The run_11 structural baseline is planar (no DEM). The promoted final contract XODR (`08_final_structural_gap.xodr`) derives from the supplementary elevation run and carries non-planar elevation (362–392 m); it is retained as supplementary evidence and does not alter the run_11 structural measurements.

3.4. Licensing and data usage

OpenStreetMap (OSM) data used in this thesis is governed by the Open Database License (ODbL), so OSM-derived datasets must retain proper attribution to OpenStreetMap contributors OpenStreetMap Foundation (2026); Open Data Commons (2009). Elevation data and auxiliary assets remain subject to their respective provider licenses. For the experiments reported here, data sources are recorded in the run configuration snapshot, and the thesis reports academic use only rather than redistribution of third-party proprietary assets.

3.5. OpenDRIVE concepts used in this thesis

ASAM OpenDRIVE models static road networks as roads with a reference line geometry and explicit lane topology ASAM e.V. (2026). This thesis focuses on a minimal but sufficient subset for CARLA ingestion:

- **`planView`:** the 2D road reference line geometry (line/arc/spiral segments).
- **`elevationProfile`:** vertical alignment as a function of longitudinal coordinate s.
- **`laneSection` and `laneLink`:** lane cross-sections and predecessor/successor connectivity that define drivable navigation.

3.6. CARLA ingestion constraints and practical failure modes

CARLA ingests OpenDRIVE via Standalone Mode (`client.generate_opendrive_world()`) CARLA Team (2026). In practice, CARLA is not tolerant to invalid topologies and may fail hard on: (i) broken lane successor/predecessor links, (ii) floating or disconnected road fragments, (iii) extremely large maps that exceed resource limits and can lead to crash-like states (e.g., "zombie" servers). Therefore, *valid OpenDRIVE* (XML/schema validity) is not necessarily *structurally validated*. The pipeline enforces stricter constraints aligned with CARLA's internal assumptions.

3.7. Sensor calibration and perception rig on the ego vehicle

Perception experiments require that simulated sensors are attached to the ego vehicle using a fixed, documented calibration. This thesis uses a `calib_data.json` configuration that specifies a multi-camera rig and a LiDAR, including image sizes and rigid-body transforms. To avoid confounds from lens distortion modeling, the cameras are treated as **ideal pinhole cameras**: we use the provided `K_undistortion` matrix for all cameras and ignore the original distortion parameters (`K` and `D`).

Camera model. Given a 3D point $X_v = [x\ y\ z\ 1]^\top$ in the ego-vehicle coordinate frame, the point in camera coordinates is obtained via the homogeneous transform

$$X_c \;=\; T_v^c\, X_v, \tag{2}$$

where T_v^c corresponds to the `cTv` matrix in the calibration file (vehicle $\to$ camera). Using a pinhole projection, the pixel coordinates $p = [u\ v\ 1]^\top$ satisfy

$$\lambda\, p \;=\; K_{\text{undist}} \begin{bmatrix} I_{3\times 3} & 0 \end{bmatrix} X_c, \tag{3}$$

with depth $\lambda > 0$ and camera intrinsics K_{undist}. This is the standard central-projection model used in multi-view geometry. Hartley and Zisserman (2004)

LiDAR extrinsics. For the LiDAR, the calibration provides a transform `vTl` from LiDAR $\to$ vehicle. A LiDAR point X_l is mapped into the ego frame as

$$X_v \;=\; T_l^v\, X_l. \tag{4}$$

Coordinate frames in CARLA. In CARLA, sensors are attached to actors using a `carla.Transform` defined relative to the parent actor (here: the ego vehicle). CARLA Team (2025a) CARLA uses a left-handed coordinate system in Unreal Engine, where the vehicle frame is typically interpreted as x forward, y right, and z up. CARLA Team (2025b) These conventions are used consistently when converting the calibration matrices into CARLA sensor transforms, ensuring that all datasets and cross-map evaluations use identical sensor geometry.

Implementation note (extrinsics direction). Because the calibration file mixes transform directions (`cTv` is vehicle → camera, while `vTl` is LiDAR → vehicle), the pipeline normalizes all transforms into a consistent convention and records the resulting 6-DoF poses in the run manifest. This prevents silent left/right inversion or axis flips, which would otherwise introduce artificial "domain gaps" unrelated to the map structure.

3.8. Derived requirements for a research-grade pipeline

From the above constraints, the thesis derives the following requirements.

- **Topological determinism:** the same input and settings must yield invariant topological signatures (road count, junction count, lane distributions); byte-level output identity is not required and is not achieved in practice (RQ1 verdict: topologically `DETERMINISTIC`, byte-level `NONDETERMINISTIC`).
- **Audit artifacts:** hash-based tracking, structured reports, and overlays per stage.
- **Offline checks:** structural correctness checks without launching CARLA.
- **Two-pass validation:** offline validation gates first, followed by CARLA execution as an empirical QA layer.

3.9. Manual map integration as a reproducibility control

The manually modeled Ingolstadt environment was provided as CARLA-compatible map assets (Grid0821 and Grid0828). Integration required ensuring that CARLA base content matches the runtime version (CARLA 0.9.16). During initial integration, a version mismatch caused deterministic crashes during world load due to incompatible blueprint bytecode (e.g., actor factory assets such as walker/vehicle factories). The issue was resolved by restoring the CARLA 0.9.16 blueprint packages from a clean reference build; after restoration, both maps loaded reliably.

This incident is reported as part of the methodology because it demonstrates experimental control: without a stable simulator state, measured differences could be artifacts of a broken runtime rather than true domain-gap effects. It is an internal governed integration observation rather than a literature-derived claim and is included here as empirical runtime provenance for the manual-map baseline.

3.10. Map provenance and comparability

Two map domains are compared under identical simulator and sensor conditions:

- **OSM-generated maps:** produced by the thesis pipeline from a fixed Ingolstadt OSM extract (deterministic settings and logged seeds).
- **Manual reference maps:** externally provided, manually modeled CARLA maps (Grid0821, Grid0828) used as reference environments.

Where simulator execution was possible, the same CARLA version and sensor configuration were used. Direct paired perception execution across both the generated and manual Ingolstadt domains was not completed.

3.11. CARLA map cache artifacts

When a map is loaded, CARLA generates and/or reuses cache artifacts that stabilize road logic and traffic behavior:

- `OpenDrive/<Map>.xodr`: the OpenDRIVE representation used by CARLA for road logic and routing.
- `TM/<Map>.bin`: Traffic Manager topology/routing cache.

These artifacts are created on first load and reused thereafter, contributing to repeatability when the simulator content is unchanged.

3.12. Failure modes and mitigation

Several operational risks affected the engineering environment around the experiments, but they were treated as controlled boundary conditions rather than as thesis results. The most important ones were simulator content/version mismatches, limited disk space for large CARLA assets, non-obvious log locations on Windows, and rendering pressure when adding sensors to tiled maps. These issues motivated explicit environment pinning, artifact logging, and low-resolution or headless fallback modes so that engineering instability would not be mistaken for a structural map-generation finding.

4. Automated OSM-to-CARLA Map Generation Pipeline

This chapter describes the automated pipeline that converts OpenStreetMap (OSM) data into CARLA-compatible OpenDRIVE (XODR) maps and spatial tiles. CARLA is a high-fidelity simulator for autonomous driving research, but its OpenDRIVE ingestion is sensitive to structural inconsistencies; therefore, the generation process must target both OpenDRIVE correctness and CARLA-specific failure modes Dosovitskiy et al. (2017); ASAM e.V. (2024). The proposed pipeline is implemented as an auditable sequence of deterministic stages with explicit contracts and persistent artifacts.

4.1. Overview and design goals

The thesis requires multiple pipeline runs and artifact variants from OSM data OpenStreetMap contributors (2026) to evaluate the structural domain gap between automatically generated maps and a manually authored reference map of Ingolstadt. The map generation process must satisfy three constraints:

- **Geometric validity** with respect to OpenDRIVE (continuous and well-formed planView and elevation profiles).
- **Semantic completeness** required for CARLA driving and sensor simulation (lanes, lane connectivity, and traffic-relevant semantics).
- **Reproducibility** to enable controlled experiments and fair comparisons across runs.

To meet these constraints, the pipeline writes a sequence of intermediate OpenDRIVE snapshots and diagnostic artifacts, enabling ablation studies and failure localization when CARLA import issues occur. Figure 1 shows the high-level stage sequence from OSM extraction to domain-gap metrics.

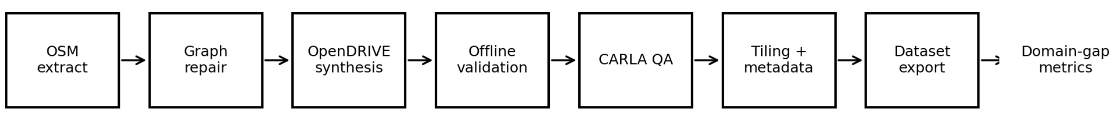

Figure 1: High-level stage sequence of the governed OSM-to-CARLA map generation pipeline. Each stage produces a versioned intermediate artifact; the final output feeds directly into the domain-gap metric computation. Source: author's own diagram.

4.2. Input data and spatial scope

The spatial scope is defined by a fixed GPS bounding box in Ingolstadt. The bounding box is used to extract the OSM region deterministically and to align downstream elevation data. The manually authored Ingolstadt OpenDRIVE map is treated as an immutable reference baseline; the pipeline generates one or more automatic variants from OSM using identical spatial bounds to isolate map-construction effects from geographic changes. Because OSM semantics are incomplete and heterogeneous, the pipeline uses conservative, deterministic rules for missing lane or junction annotations and logs all inferred decisions as artifacts.

4.3. Coordinate reference handling and canonical evaluation frame

A recurring source of invalid structural gap measurements is inconsistent georeferencing across OpenDRIVE sources. The manual Ingolstadt reference maps (Grid0821/0828) are provided in a Transverse Mercator projection consistent with the UTM Zone 32N family (metric coordinates in meters), while automatically generated OSM-derived maps may use a local Transverse Mercator projection anchored to the project bounding box and additionally encode a global translation via the `<offset>` element. Directly comparing raw `(x,y)` coordinates across these frames can therefore produce kilometer-scale errors that are unrelated to actual map differences.

To ensure meaningful structural metrics, this thesis adopts a *canonical evaluation frame*: all geometric operations (alignment, distance metrics, and tile bounding boxes) are performed after transforming map geometry into EPSG:32632 (WGS84 / UTM Zone 32N), with `<offset>` applied before projection. The transformation is implemented via deterministic header parsing (including CDATA and whitespace normalization) and is audited via a georeference provenance record written per run. After canonicalization, alignment is restricted to rigid SE(2) transforms (rotation + translation, scale fixed to 1) and guarded by a sanity threshold to prevent reporting misleading metrics when overlap is insufficient.

Figure 2 summarizes the normalization and alignment workflow.

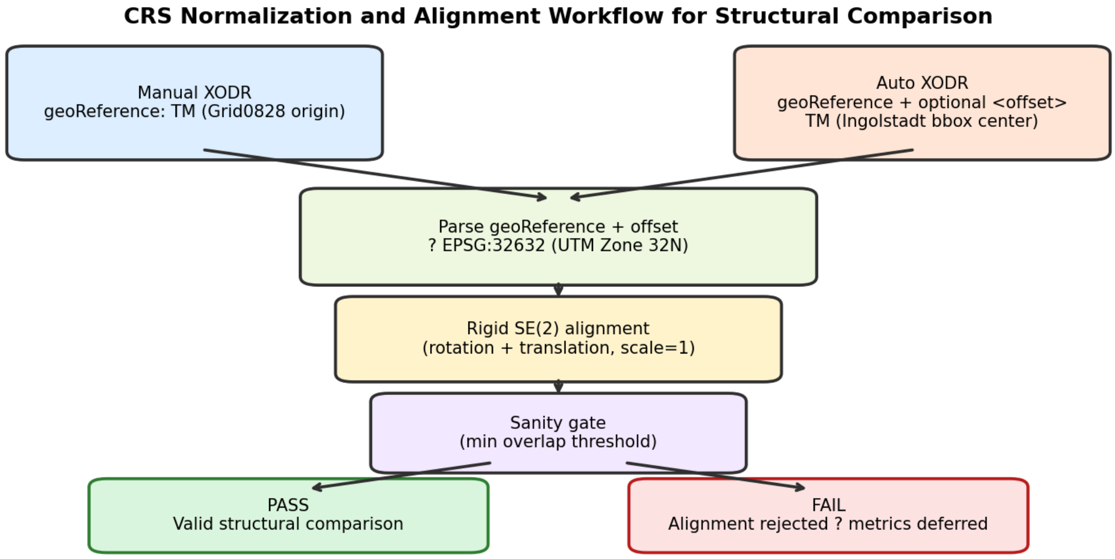

Figure 2: Coordinate normalization used for valid comparison between manual and automatically generated OpenDRIVE maps. Both sources are parsed (geoReference and optional offset), projected to a canonical CRS (EPSG:32632), and then aligned using a rigid SE(2) transform with a sanity gate.

4.4. Stage-by-stage map construction

The pipeline is organized into sequential stages. Each stage writes a versioned OpenDRIVE output (XODR) and a structured JSON report so that the full transformation chain remains traceable from input to final OpenDRIVE output.

4.4.1. Sanitation (Stage 1)

The sanitation stage normalizes OpenDRIVE metadata (e.g., `geoReference`) and repairs common formatting issues that can prevent downstream tools from parsing the map. It also generates a cropped "QA slice" for rapid inspection. The stage output serves as a stable baseline for subsequent topology and geometry processing.

4.4.2. Topology lint and optional SUMO repair (Stage 2)

Automatic conversion from OSM frequently produces junction artifacts, missing connections, and inconsistent road-link structures. The pipeline therefore performs a topology lint pass and can optionally execute a SUMO-based repair workflow to stabilize the road graph Behrisch et al. (2011b). A validation report is stored as machine-readable JSON to support reproducible error analysis and comparisons between pipeline versions.

4.4.3. Topology repair (Stage 3)

Stage 3 applies deterministic cleanup of road predecessor/successor relationships and junction references. The goal is to ensure that the connectivity graph is well-formed before higher-level semantic enrichment is applied. The current governed pipeline also canonicalizes junction connectors before geometry freeze: true duplicate connector families are collapsed by `incomingRoad`/`exitRoad`, and irregular same-start connector bundles may have their `planView` geometry normalized without deleting distinct turn semantics. The pass writes a Stage 3 canonicalization report and enforces invariants so that no incoming road loses all valid junction connectivity.

4.4.4. Enrichment and object injection (Stage 4)

To support perception studies and qualitative inspection, the pipeline optionally inserts semantic objects and environmental structure. Depending on data availability, this includes buildings derived from OSM footprints and, when available, preprocessed building footprints provided as GeoJSON Butler et al. (2016), as well as additional realism objects. Enrichment is treated as an additive overlay: it is not allowed to mutate plan-view geometry or lane topology and is later verified by integrity gates.

An auxiliary 3D enrichment path based on OSM2World Knerr et al. (2024) was integrated for mesh-level scene generation. OSM2World converts the OSM source to an OBJ scene file Knerr et al. (2024); a stack-check conversion for the Ingolstadt area (using a validated OSM extract, SHA-256 confirmed) produced an approximately 398 MB `scene.obj` artifact in approximately 252 seconds. A subsequent Blender Blender Foundation (2024)-driven OBJ→FBX stage converts the mesh to the format required for Unreal Engine asset cooking, enabling optional import of realistic environmental geometry (buildings, vegetation) into a packaged CARLA world. Both stages use deterministic caching keyed on SHA-256 hashes of the input OSM file and configuration, writing a `osm2world_status.json` or `blender_status.json` provenance record. This auxiliary path is disabled by default (`ENABLE_OSM2WORLD=0`) and does not affect OpenDRIVE generation. Importing OSM2World-derived FBX assets into a distributable CARLA world additionally requires the full Unreal Engine source build of CARLA, which is outside the scope of this thesis; only the XODR-based standalone API (`generate_opendrive_world`) was used for simulator-dependent evaluation here. The structural domain-gap analysis in this thesis is therefore performed on the final OpenDRIVE road-network artifact, and OSM2World-derived mesh enrichments lie outside the primary comparison scope.

4.4.5. Elevation application from DEM (Stage 5)

Because the pipeline supports DEM-backed 3D-capable artifacts, elevation is implemented as a first-class pipeline capability rather than a cosmetic detail. A digital elevation model (DEM) aligned to the GPS bounds can be sampled along roads and encoded in OpenDRIVE elevation profiles, and a smoothing operation is applied to avoid unrealistic high-frequency elevation changes that can break collision meshes or produce implausible slopes. The elevation source in this thesis is a Copernicus DEM product retrieved via OpenTopography, which provides reproducible access to global DEM datasets and metadata for scientific workflows OpenTopography (2025); Copernicus Programme (2021). For the authoritative structural analysis discussed in Chapter 7, however, the run_11 comparison artifact remained flat at $z = 0$ even though later governed artifacts proved that DEM-backed generation is possible. The reported primary structural gap metrics are therefore planar (2D) only, and Section 7, especially the paragraph "Elevation status and CARLA visual QA boundary," should be read as the authoritative interpretation for the structural analysis artifact.

4.4.6. PlanView and continuity smoothing (Stage 6)

OSM-derived geometry often contains fragmented plan-view elements and heading discontinuities. Stage 6 merges micro-segments, smooths plan-view geometry, clamps pathological curvature, and performs continuity repair to reduce geometric noise while preserving the global road layout. This stage typically reduces the number of geometric primitives and improves simulator stability by preventing discontinuities that trigger CARLA import failures.

4.4.7. Lane and cross-section generation (Stage 7)

For CARLA driving simulation and dataset generation, lane semantics must be present and consistent. This stage generates or repairs lane structures, applies lane-offset smoothing, and enforces cross-section continuity constraints to prevent negative widths and discontinuities at laneSection boundaries. A lane overlay visualization and cross-section diagnostics are exported for manual plausibility checks.

4.4.8. Lane links, markings, and integrity gates (Stage 8)

The final stage creates or repairs lane-to-lane connectivity, generates lane markings, and enforces CARLA-specific successor invariants at the laneSection level. Additional integrity gates (e.g., collision-mesh plausibility checks) detect major polygon issues that are strongly correlated with CARLA import failures. The output of this stage is the *contract-checked* OpenDRIVE file used in subsequent experiments. After late junction-link mutation, the selected final XODR is rechecked by lane-connectivity, junction-integrity, and connector-crowding gates. Conservative connector pruning and the `junction_connector_risk.json` audit are part of this finalization path; if residual connector crowding exceeds configured thresholds, promotion fails rather than silently accepting the artifact.

4.5. Quality gates and integrity contracts

Gate Name	Condition	Action on Failure	Protects Against
Min XODR file size	$\geq$ 10,000 bytes	Hard stop	Stub fixture inputs (run_02: 796 bytes)
Tile IoU gate	IoU ≥ 0.5	Skip per-tile metrics	Invalid local comparisons
CRS sanity gate	Overlap after alignment > threshold	Reject, report deferred	CRS mis-projection artefacts
Empty correspondence gate	`correspondence.csv` $\geq$ 1 data row	Hard stop	Silent zero-row claims
Claim boundary annotation	`_claim_boundary` in `full_report.json`	Required for admissibility	Unannotated result propagation
Connector-crowding gate	Repeated-anchor and overlap counts within thresholds	Hard stop	Exploded junction patches / connector crowding

Table 3: Thesis integrity gates applied to prevent false-positive structural claims.

Table 3 summarizes the thesis-specific gate conditions that prevent broken correspondence, CRS mismatch, or stub artifacts from being misreported as valid structural evidence.

The run_02 stub defect (796-byte XODR) was retroactively blocked by Gate 1 after discovery (Table 3); run_03 was the first re-run with real inputs (15.3 MB auto, 65.9 MB manual XODR), but the final authoritative structural thesis metrics are reported from the governed run_11 raw basis (Section 7).

All gate decisions, override flags, and bypass history are recorded in a machine-readable `kill_switch_provenance` block within `full_report.json`. Both the identity-alignment guard and the empty-correspondence guard are enforced by default; they can only be bypassed by explicitly setting `UP_ALLOW_IDENTITY_ALIGNMENT=1` or `UP_ALLOW_EMPTY_CORRESPONDENCE=1` respectively, and any such override is permanently logged in the provenance block. Domain-gap metrics are normalized to $[0, 1]$ using explicit reference scales drawn from `SETTINGS.DOMAIN_GAP_NORMALIZATION`: geometry RMSE reference = 1.0 m, Hausdorff reference = 5.0 m, curvature KL reference = 0.5, intersection delta reference = 1.0, semantic delta reference = 1.0. The effective normalization applied to each metric is reported verbatim in the `normalization_contract` block of `full_report.json`.

4.6. Outputs and reproducibility artifacts

Each pipeline run produces (i) a sequence of stage XODR outputs, (ii) per-stage preview images for qualitative verification, and (iii) diagnostic JSON files that quantify structural properties (e.g., road counts, junction counts, curvature statistics) and validation outcomes. A settings snapshot is written to record all run-defining parameters; combined with cryptographic hashes of key artifacts, this snapshot enables reproducible re-runs and supports determinism auditing.

4.7. Design and reproducibility principles

The governed pipeline follows five principles: deterministic execution under pinned inputs, a single geometry authority for `planView` and `elevationProfile`, artifact-first logging, offline-first validation, and causal separation between synthesis and evaluation. In practice, this means that structural evidence must remain interpretable even when CARLA is unavailable or unstable, and that later perception-facing steps are not allowed to retroactively alter the structural analysis basis.

4.8. Artifact contract and determinism reporting

Each run preserves a compact artifact contract rather than only a final XODR file. At minimum, the governed run directory contains the input snapshot, stage outputs, key diagnostic JSON files, and run-defining hashes. This contract serves two purposes. First, it allows later stages to be audited without re-running the full conversion chain. Second, it distinguishes byte-level file drift from changes that would matter scientifically, such as altered road counts, junction counts, or reference-line geometry.

Determinism is therefore reported at the artifact level, not inferred from a single successful load. Repeated runs are compared through settings snapshots, hashes, and semantic signatures so that structural invariance can be defended even when XML serialization varies across runs Peng (2011); Sandve et al. (2013).

4.9. Summary

At Bachelor-thesis scope, the important point is not every implementation detail of the governed code base, but the staged logic of the conversion path: OSM input is sanitized, topology is repaired, semantics are enriched, geometry is stabilized, lanes and lane links are repaired, and the result is accepted only if explicit integrity gates are satisfied. This staged view is the basis for the structural experiments in Chapters 5–7 and for the simulator-dependent QA discussed in Chapter 6.

5. Map Generation and Repair Method

This chapter describes the deterministic pipeline stages that transform an OSM extract into structurally validated OpenDRIVE tiles. The goal is not to maximize semantic completeness, but to enforce a set of invariants that make city-scale ingestion repeatable and auditable ASAM e.V. (2026); CARLA Team (2026).

5.1. Input preprocessing and sanitization

OSM extraction and caching. The pipeline operates on a bounded geographic region (Table 1). The OSM extract is stored locally and reused across runs to avoid nondeterminism from live downloads.

Coordinate normalization. OSM coordinates are provided in WGS84 latitude/longitude. Geometry is projected to a local metric frame to enable robust distance-based operations (clustering, smoothing) and OpenDRIVE synthesis. The chosen projection parameters are recorded in the settings snapshot.

5.2. Per-tile correspondence and coverage

Whole-map metrics can obscure where structural discrepancies are concentrated. For this reason, the evaluation pipeline supports per-tile matching and tile-local gap metrics. However, per-tile comparisons are only valid if both maps are partitioned using the *same* tile grid definition: identical projected CRS (meters), identical grid origin, identical tile size, and identical buffer rules. If these conditions are not met (e.g., manual tiles missing or computed in a different frame), the pipeline explicitly skips per-tile metrics and writes diagnostics (e.g., `tile_correspondence.csv`) rather than fabricating matches.

In practice, automatic maps are tiled as part of the generation process, while manual reference maps must be tiled with the same grid parameters derived from the auto run metadata (origin and tile size). Figure 3 illustrates the correspondence requirement. This conservative gating prevents invalid local claims when overlap is partial or coordinate frames are inconsistent.

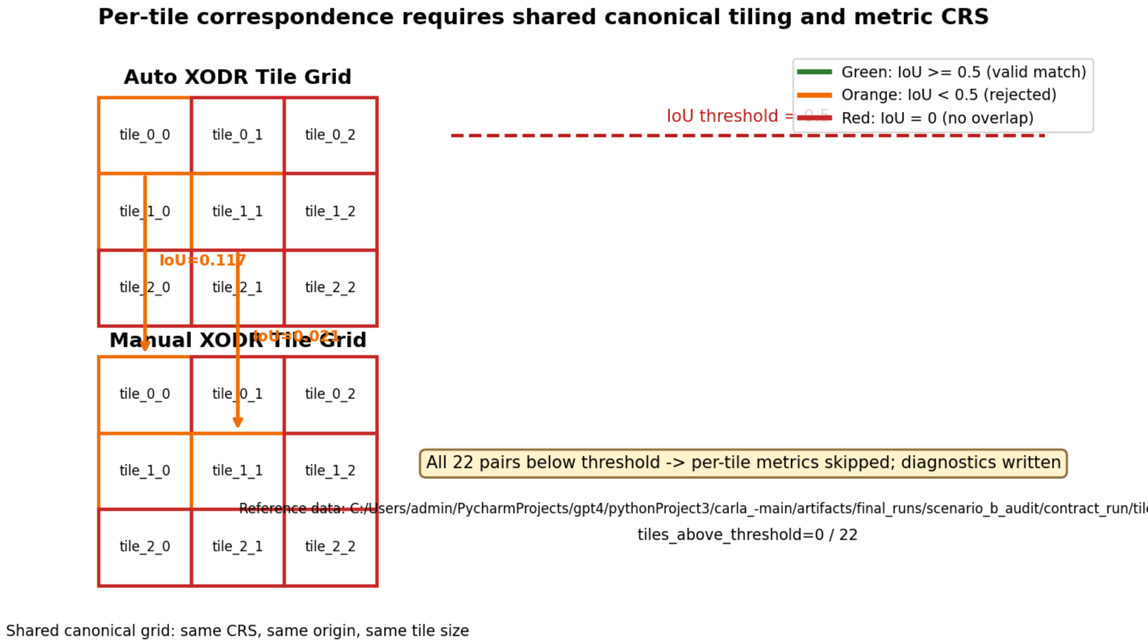

Figure 3: Per-tile correspondence requires a shared canonical tile grid (origin and tile size) and a consistent metric CRS. Manual and auto maps are tiled using identical grid parameters; pairing is performed by tile bounding-box IoU with explicit diagnostics.

5.3. Operational definition of drivability

During pipeline development, a central engineering requirement was that generated maps should support a minimal CARLA import and spawn workflow. To avoid an informal interpretation, the term *drivability* is used here only in this narrow engineering sense and not as evidence of visual usability, route-level stability, or perception readiness. A map (or tile) is considered drivable for the purposes of this section if it satisfies all *offline* structural invariants and passes a minimal *simulator-dependent* import and spawn test.

Optional higher-level drivability tests (e.g., autopilot driving for a fixed horizon) are treated as *scenario* validation and are discussed separately from the minimal import/spawn contract.

Negative-s violations are repaired automatically in-place during the tiling stage (Stage 9): each affected tile is first backed up, then the violation is corrected, and a per-tile report (`xodr_negative_s_report.csv`) records before-and-after violation counts across 11 OpenDRIVE XPath targets (geometry, lane sections, lane offsets, elevation profiles, superelevation, lane widths, lane borders, road objects, signals, road type, and speed entries). Monotonic-sequence violations are reported but not automatically repaired, as they require semantic interpretation of the road layout to resolve safely.

Table 4: Operational drivability criteria and where they are verified.

Criterion	**Verified by**	**Evidence artifact**
OpenDRIVE XML well-formed and schema-consistent	Offline quality gates	`validation_report_full.json`
No critical parameterization violations (e.g., non-negative s in laneSections)	Offline quality gates + repair report	`xodr_negative_s_report.csv`
Lane topology usable for routing (successor/predecessor links resolved)	Offline topology gates	`gate_failures.json`
CARLA can import the OpenDRIVE and generate the world	Tile QA worker (CARLA)	`tile_qa_reports/*.json`
At least one valid spawn point for the ego vehicle	Tile QA worker (CARLA)	`tile_failure_taxonomy.json`

5.4. Topology linting and structural risk scan

Before repairs, diagnostic scans quantify risks: connected components, near-miss endpoints, anomalous curvature outliers, and missing junction references. High-risk road identifiers are recorded to enable targeted debugging rather than opaque global modifications.

5.5. Topology repair and enrichment

Intersection clustering. Near-miss endpoints are clustered within a deterministic tolerance. Cluster merges are logged. Layer information (bridge/tunnel/layer tags) is used to avoid illegal merges across vertical separation when available.

Semantic enrichment. Traffic lights and simple scene elements can be inferred and inserted conservatively to support more diverse perception scenes. Enrichment is designed to be *geometry-preserving*.

5.6. Geometry authority: planView smoothing and elevation

OpenDRIVE uses a reference line and longitudinal coordinate s to define road geometry ASAM e.V. (2026). The pipeline constructs the reference line and then stabilizes it via deterministic smoothing and continuity repair.

Elevation integration. A DEM is sampled along the reference line and encoded as `elevationProfile`. To reduce high-frequency artifacts, elevation is smoothed while preserving large-scale terrain trends.

Geometry freeze policy. After continuity stabilization, geometry is declared frozen: downstream stages may not modify `planView` or `elevationProfile`. This prevents feedback loops where lane fixes reintroduce geometric defects.

5.7. Lane generation and cross-section repair

Lane sections are derived from OSM tags where available (e.g., lane counts) and from deterministic defaults otherwise Haklay (2010). Cross-section repair enforces positive widths and continuity across adjacent lane sections. Sidewalks may be added as additional lane types.

5.8. LaneLink construction and the CARLA successor invariant

CARLA routing stability depends on consistent lane predecessor/successor relationships CARLA Team (2026). The pipeline enforces a strict successor invariant for drivable lanes.

Invariant definition. Let $G = (V, E)$ be the directed lane graph where V are lanes and $(u, v) \in E$ represents a successor link from lane u to lane v. For every lane u marked as drivable, the successor set must be non-empty:

$$\forall u \in V_{\text{drive}} : \quad \exists v \in V_{\text{drive}} \text{ such that } (u, v) \in E. \tag{5}$$

Violations are repaired where possible (e.g., by connecting to the best matching downstream lane) or the lane is downgraded to a non-driving type to prevent CARLA failures.

5.9. Tiling and adjacency graph generation

City-scale maps are decomposed into spatial tiles to localize errors, speed up validation, and support scalable experiments. Tiles are extracted with a configurable buffer to preserve cross-boundary successors. Figure 4 shows the buffered extraction scheme used to reduce cross-tile successor leakage.

Each tile export includes bounds, road/lane counts, successor leakage statistics, and hashes for reproducibility. An adjacency graph encodes tile neighborhood relations for later analysis and visualization.

5.10. Seam continuity evaluation between adjacent tiles

Tiling introduces artificial boundaries that can create discontinuities in an otherwise globally consistent road network. Such seam artifacts may not prevent simulator loading, yet they can

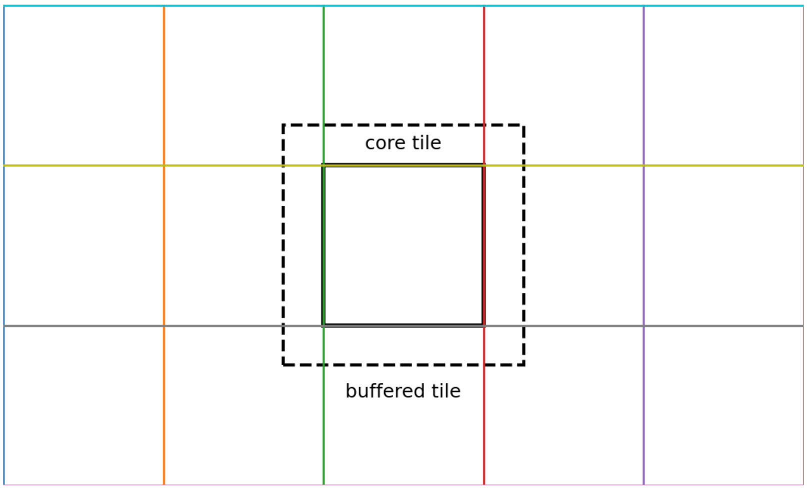

Figure 4: Buffered tiling concept. The dashed region denotes the buffered extraction used to reduce successor leakage across tile boundaries.

affect downstream tasks: routing continuity, vehicle dynamics at boundary crossings, and the visual/geometry consistency of perception datasets.

Seam metrics. For each pair of adjacent tiles (A, B), the pipeline evaluates seam continuity by comparing the boundary-near road geometry and lane centerlines on both sides. Three metrics are recorded:

- **Lateral displacement** $e_\perp$ (meters): planar offset between matched boundary samples.
- **Heading discontinuity** e_ψ (degrees): difference in tangent heading, wrapped to $[-180°, 180°]$.
- **Elevation jump** e_z (meters): difference in elevation at the seam.

Conceptually, the comparison matches the "end" of a boundary-crossing primitive in tile A to the corresponding "start" primitive in tile B (after applying the same tiling buffer and coordinate frame).

Adjacency-complete evaluation. To avoid bias from traversal order (e.g., "previous tile loaded"), seam evaluation iterates *all* edges in the stored tile adjacency graph. This enables reporting seam statistics per adjacent tile pair and supports reproducible aggregation (median/95th/max) over a fixed edge set.

Failure mode: empty adjacency graph ("0/0 edges"). A key robustness issue during development was that the stored adjacency graph could contain zero edges (e.g., "0 edges for 74 tiles"). In this situation, seam checks cannot execute and the seam stage would otherwise report "0/0 edges," which is diagnostically equivalent to "seam QA never ran." In practice, the root cause was a mismatch between the produced tile set and the adjacency consumer (e.g., schema differences or inconsistent parsing of tile identifiers from filenames).

Deterministic fallback adjacency reconstruction. To guarantee that seam analysis executes on multi-tile runs, the seam stage includes a deterministic fallback. If an adjacency file exists but yields zero edges while multiple tiles are present, the pipeline reconstructs a 4-neighbor grid graph directly from tile filenames with encoded integer indices (e.g., `tile_3_7.xodr`). For each tile index pair (i, j), neighbor edges are added to $(i \pm 1, j)$ and $(i, j \pm 1)$ if the corresponding tiles exist. The parsed index set and the resulting edge count are logged, and a `..._fallback.json` adjacency artifact is written for reproducibility. If adjacency remains empty despite a multi-tile set, the QA stage fails fast with an explicit error rather than silently skipping seam checks.

Reproducible seam artifacts. Each evaluated edge (A, B) produces a machine-readable seam report (JSON) containing: tile identifiers, hashes of the compared OpenDRIVE files, parameters (tile size, buffer, sampling step), and the resulting seam metrics with warnings. A deterministic manifest lists all seam reports in stable order and links to the run-wide settings snapshot. This makes seam results externally auditable and reproducible without re-running CARLA.

5.11. Quality gates

Because an OpenStreetMap-to-OpenDRIVE conversion can fail silently or produce partially valid outputs, the pipeline uses a staged *quality gate* strategy. Gates are designed to (i) detect invalid or simulator-hostile structures early, (ii) record a failure taxonomy for later analysis, and (iii) prevent confounding of domain-gap experiments by broken environments.

Offline gates (structural and semantic sanity). Offline gates operate directly on OpenDRIVE and derived geometry without requiring CARLA. They include:

- **XML/schema integrity:** well-formed OpenDRIVE and required elements present.
- **Parameterization invariants:** consistency of longitudinal parameter s (e.g., no negative `laneSection@s`); consistency between geometry lengths and `road@length`.
- **Topology invariants:** resolvable successor/predecessor relations and junction connectivity sanity.
- **Geometry feasibility:** thresholds on extreme curvature, micro-segment merging and continuity checks after smoothing.

- **Semantic sanity:** a semantic-overlap check (e.g., mutually exclusive road-space labels) and a randomness/entropy check used to detect degenerate or overly repetitive map realizations.
- **Collision mesh gate:** polygon validity checks for generated road-space meshes (implemented using robust computational geometry).

All gate outcomes are recorded in a structured report (e.g., `validation_report_full.json`) and a concise failure list (e.g., `gate_failures.json`), enabling reproducible post-hoc analysis.

Repository-specific validity gates (elevation seams and origin sanity). In addition to generic OpenDRIVE invariants, the implementation includes two thesis-critical gates motivated by CARLA failure modes observed on large, tiled maps: (i) an **elevation seam gate** that detects large discontinuities in the road surface between adjacent tiles (which otherwise appear in CARLA as vertical "walls"/trenches and can destabilize physics), and (ii) an **origin sanity gate** that rejects maps whose coordinate frame is absurdly far from the simulator origin (a common symptom of projection mistakes). Gate outcomes are summarized into a compact `map_acceptance.json` schema (`valid`, `failed_gates`, `summary`). For strict experimental hygiene, perception capture can be configured to require acceptance (e.g., skipping runs on invalid maps rather than silently mixing corrupted data into the training set).

Map identity guarding (manual vs. generated). Because the thesis compares a manually authored Ingolstadt map against many generated realizations, the CARLA-side tooling includes a *map identity guard* that records and verifies the loaded world/map name before spawning sensors or logging data. This prevents accidental cross-contamination of results when running headless batches or recovering from simulator restarts.

Simulator-dependent gates (CARLA import and spawn). CARLA-based checks are expensive and may be unstable for large maps. Therefore, they are executed after offline gates and typically on spatial tiles. For each tile, the validation worker attempts to import the OpenDRIVE into CARLA, generate the world, and spawn an ego vehicle with the fixed sensor rig. Outcomes are summarized per tile (JSON) and aggregated into a failure taxonomy (e.g., `tile_failure_taxonomy.json`), allowing pass-rate and failure-mode statistics to be reported separately from map correctness.

Relation to domain-gap experiments. Only maps/tiles that satisfy the drivability contract in Table 4 are used for sensor-based perceptual evaluation and for training data generation. Maps failing gates remain informative: failure rates and repair counts are reported as *pipeline stability metrics*, but they are not mixed into perception results to avoid bias.

5.12. Alignment and comparison protocol

After map generation, the auto and manual OpenDRIVE files are brought into a common metric reference frame before any distance metric is interpreted. The comparison first applies header-based CRS normalization and then estimates a rigid SE(2) alignment with scale fixed to one. This preserves the physical meaning of road lengths and prevents artificial scale shrinkage from lowering geometric error.

Per-tile metrics are only reported if a valid spatial correspondence can be established. When correspondence is empty or frame compatibility cannot be justified, the governed path refuses per-tile interpretation and falls back to whole-network structural reporting. This refusal logic is important methodologically because it prevents localized numbers from being reported on mismatched spatial partitions.

5.13. Structural metrics

The thesis reports complementary metrics because no single number captures the full structural gap.

Whole-network RMSE. Let $\{\mathbf{p}_i\}_{i=1}^{N}$ denote sampled points from the candidate map and let $\{\mathbf{q}_i\}_{i=1}^{N}$ be their closest points on the manual reference after rigid alignment. The whole-network RMSE is

$$\mathrm{RMSE}_{\mathrm{whole}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\|\mathbf{p}_i - \mathbf{q}_i\|_2^2}. \tag{6}$$

This metric is sensitive to coverage mismatch because roads that exist only in one map still contribute large distances.

Matched-subset RMSE and match rate. To separate local alignment quality from coverage surplus, the pipeline also reports RMSE only on the subset of road samples that have a counterpart within a fixed acceptance threshold $\tau = 50\,\mathrm{m}$:

$$\mathrm{RMSE}_{\mathrm{matched}} = \sqrt{\frac{1}{|M|}\sum_{i\in M}\|\mathbf{p}_i - \mathbf{q}_i\|_2^2}, \tag{7}$$

where $M = \{i \mid \|\mathbf{p}_i - \mathbf{q}_i\|_2 \leq \tau\}$. The match rate is

$$\text{match rate} = \frac{|M|}{N}. \tag{8}$$

Together, the matched-subset RMSE and the match rate indicate how much of the manual network is represented locally well by the generated map.

Hausdorff distance. Worst-case geometric deviation is captured by the symmetric Hausdorff distance:

$$d_H(P, Q) = \max\left(\max_{p\in P} \min_{q\in Q} \|p - q\|_2,\ \max_{q\in Q} \min_{p\in P} \|p - q\|_2\right). \tag{9}$$

Unlike RMSE, this metric is dominated by the most extreme structural outlier and therefore exposes coverage failures and isolated geometry defects.

Bounding-box IoU. Because kilometer-scale errors can arise from CRS mismatch before any meaningful alignment is attempted, the pipeline also reports the intersection-over-union of the aligned plan-view bounding boxes:

$$\mathrm{IoU}_{\mathrm{bbox}} = \frac{\mathrm{area}(B_{\mathrm{auto}} \cap B_{\mathrm{manual}})}{\mathrm{area}(B_{\mathrm{auto}} \cup B_{\mathrm{manual}})}. \tag{10}$$

This is not a road-shape metric; it is a coarse sanity check that the two maps occupy the same spatial frame after normalization.

Curvature and companion descriptors. Curvature divergence, road-length ratio, junction ratio, and connectivity descriptors complement the distance metrics. Their role is explanatory: they help identify whether the gap is dominated by local geometry noise, global coverage surplus, or topological encoding differences.

5.14. Perception protocol boundary and sensor-rig consistency

Perception-related claims are made only under a single authoritative rig contract. Camera intrinsics use the undistorted pinhole matrix from `K_undistortion`; camera extrinsics apply `cTv` directly; LiDAR extrinsics invert `vTl` before attachment. These conventions are fixed so that any reported sensor differences can be attributed to the map rather than to accidental calibration drift.

The simulator-backed perception protocol is deliberately narrower than the structural protocol. Structural analysis can be executed offline on OpenDRIVE artifacts alone, but perceptual-gap analysis additionally requires a stable simulator world, valid ego spawn, and functioning sensor callbacks. This distinction is central to the thesis claim boundary: loadability and rig correctness are necessary preconditions, whereas completed paired capture is required for direct perceptual-gap measurement. CARLA runtime loadability was treated as a necessary technical acceptance criterion, but not as sufficient evidence of visual, geometric, or semantic correctness. In the final governed evaluation, the promoted generated XODR could be parsed and loaded in CARLA, yet visual QA still indicated observed road-mesh and surface-continuity defects. A controlled planar ablation also showed that flattening elevation did not remove these defects, indicating that the dominant failure mode is horizontal/topological rather than DEM-first (see Section 6, especially Table 5).

5.15. Reproducibility and evidence integrity

Each governed run preserves the settings snapshot, artifact hashes, and machine-readable status files needed to audit the claim boundary afterward. Failure is treated as data rather than hidden noise: empty correspondence, invalid alignment, or incomplete perception capture are recorded explicitly and downgrade the admissible claim rather than being silently ignored. This is why the thesis can report structural results confidently while still treating generated-map perception evidence as partial.

5.16. Summary

The methodology combines deterministic map generation with guarded comparison rules. Whole-network RMSE, matched-subset RMSE, match rate, Hausdorff distance, and bounding-box IoU jointly characterize the structural gap, while strict evidence contracts prevent simulator-facing failures from being misreported as valid perceptual results.

6. Simulator-Based QA and CARLA Visual Assessment

Simulator-based checks are treated as a separate evidence layer from the OpenDRIVE-only structural analysis. The governed pipeline therefore distinguishes four questions that are often conflated in simulator projects: (i) can the XODR be parsed, (ii) can CARLA load it as a world, (iii) is the rendered road mesh visually usable, and (iv) is the world stable enough for perception capture. Only the first two questions are answered positively for the final generated Ingolstadt artifact.

6.1. Offline validity versus simulator validity

Offline gates establish whether the generated OpenDRIVE is structurally interpretable: XML integrity, geometry sanity, lane-link consistency, and final artifact provenance are all checked before CARLA is involved. CARLA then acts as an empirical QA layer rather than as the sole definition of correctness CARLA Team (2026). This separation is important because a world can load despite severe road-mesh defects, and conversely a structurally meaningful artifact can still hit simulator-specific runtime boundaries.

6.2. CARLA loadability and map identity

The governed final artifact (`08_final_structural_gap.xodr`, `scenario_b_audit` contract run)[1] loads in CARLA 0.9.16 standalone mode as `Carla/Maps/OpenDriveMap`. The raw-load probe reports `load_ok=true`, `spawn_point_count=8535`, and a successful `world.get_map().to_opendrive()` export. For context, the manually-authored Grid0821/Grid0828 reference map reports approximately 11,600 spawn points in the same CARLA version; the generated artifact's 8,267–8,535 points (71–73 % of manual) is consistent with sparser junction-centric topology in which many minor residential roads carry no valid spawn locations. These checks show that the artifact is parseable and loadable, and they confirm that later diagnostic screenshots were taken from the same promoted XODR. They do *not* establish that the generated road mesh is visually usable.

6.3. Controlled planar ablation of the generated road mesh

To separate vertical DEM effects from base topology defects, the final promoted XODR was tested in three forms: the original DEM-backed artifact, a flat-elevation-only copy, and a flat-elevation-plus-lateral-profile copy. In the flat variants, all roads were forced to $z = 0$ while preserving plan-view geometry, junction topology, lanes, signals, and objects.

Table 5 summarises the core outcome: parsing and loading survive flattening, but all three tested variants continue to exhibit road-mesh and surface-continuity defects.

[1] Full artifact path: `artifacts/final_runs/scenario_b_audit/contract_run/08_final_structural_gap.xodr`

Variant	CARLA load	Visual QA	Interpretation
Original DEM-backed artifact	PASS	FAIL	Detached slabs, malformed junction patches, broken road-surface continuity
Flat elevation only	PASS	FAIL	Flattening removes terrain height, but the same defect pattern remains
Flat elevation + lateral removal	PASS	FAIL	Removing lateral profile does not resolve the broken road mesh

Table 5: Controlled planar ablation for the promoted generated XODR. CARLA load success is preserved in all variants, but visual failure remains in all variants.

The static audits reinforce the visual result. After flattening, the dominant structural issue counts remain unchanged: `planview_issue_count=3414`, `junction_issue_count=3720`, `link_gap_issue_count=1645`, and `graph_component_count=9`. `planview_issue_count` counts within-road geometry-chain discontinuities, where consecutive `planView` geometry elements fail to connect spatially within a 0.5 m tolerance; `junction_issue_count` tallies connector-road pose mismatches (start or end pose displaced from the target through-road endpoint); `link_gap_issue_count` tallies road-to-road link pairs whose declared endpoint distance exceeds a merge threshold. All three counts remain invariant after elevation removal, confirming that the dominant failure class is horizontal topology, not DEM height. The largest linked-road XY gaps remain above 1.1 km, while the corresponding Z gaps collapse to zero. The failure mode is therefore not primarily vertical. In other words, the broken slabs are not explained by DEM height alone; the stronger explanation is base topology, junction construction, road-link consistency, clipping/tiling side effects, or CARLA importer sensitivity to those horizontal defects.

6.4. Perception-readiness boundary

This visual QA result changes the safe interpretation of generated-map perception readiness. The generated XODR is loadable, but it is not visually stable enough to serve as a convincing perception world. Town10HD remains useful only as bounded evidence that the governed sensor rig, callback handling, and dataset export path work on a stable built-in CARLA map, but it cannot be used as evidence that the generated Ingolstadt map is perception-ready.

Accordingly, the thesis treats CARLA load success as a necessary but insufficient quality gate. Generated-map perceptual evidence remains bounded for two reasons: the generated road mesh fails visual QA, and the manual cooked Grid0828 capture path was additionally limited by a CARLA runtime boundary during live sensor spawn on the custom world. Direct manual-versus-generated perception comparison is therefore deferred. A minimal-rig variant using a single forward-facing RGB camera was also attempted on both Grid0821 and Grid0828; both attempts terminated with the same streaming-transport error, confirming that the failure is map-level rather than sensor-rig-complexity-dependent.[2] The failed 50-frame strict-evidence capture and the

[2] Full diagnostic evidence and root-cause analysis for the Grid0821/Grid0828 sensor-spawn boundary: `docs/submission/perception_rca_final.md` in the accompanying repository.

later rerun3 supplementary sensor-I/O probe are separate evidence layers; the latter confirms only short sensor callback functionality and is not interpreted as perception readiness. A subsequent attempt using the drivability-fixed artifact (4-pass repair, ego drive 46.45 m, Section 6.7) in CARLA standalone mode succeeded with 20 RGB frames and 20 semantic-segmentation frames captured on the generated Ingolstadt map, confirming that the drivability repair resolves the spawn-point and ego-drive limitation; however, a route-matched Town10HD counterpart capture was not completed in the same session (8 partial frames, record_route timeout), so the direct paired manual-versus-generated perceptual comparison remains deferred.

6.5. Summary

The simulator-dependent evidence supports a narrow claim: the final generated XODR is parseable and loadable in CARLA standalone mode. It does *not* support a claim of visual usability. The controlled planar ablation shows that flattening elevation does not remove the observed road-mesh and surface-continuity defects, so the dominant failure mode is horizontal/topological rather than DEM-first. This is why the structural OpenDRIVE comparison remains valid while positive generated-map visual evidence and generated-map perception evidence remain deferred.

6.6. Visual Inspection of the Automatically Generated Map

Following application of the planview-smoother anchor-guard (terminal-anchor guard, commit `1b6956e3`) and stage-08 planview gate patches (commit `a2aa2c25`), the pipeline was rerun against the Ingolstadt OSM input (rerun3, 2026-04-27). The patched map artifact was loaded in CARLA 0.9.16 via the OpenDRIVE standalone mode to confirm loadability and to obtain representative screenshots for diagnostic visual inspection. Thirty NPC vehicles were spawned via the CARLA traffic manager and an ego vehicle with a forward-facing RGB camera was driven on autopilot to capture dynamic sensor frames. This enriched probe is retained as supplementary evidence only and does not replace the authoritative run_11 structural pair. Figures 5–7 show top-down, street-level-with-traffic, and junction overhead views from the enriched probe run. All screenshots were captured by the author on 2026-04-27; the probe reported `load_ok=true` and `spawn_point_count=8267`. The images document CARLA loadability and allow qualitative inspection of road and junction geometry, but they do not establish visual usability, route drivability, or perception readiness.

6.7. Post-Submission Drivability Fix and Ego Driving Verification

After thesis submission, the final generated Ingolstadt XODR was subjected to a supplementary drivability-oriented repair pass. The motivation was practical rather than evidential: the pre-fix artifact loaded in CARLA but failed visual QA, with `planview_issue_count=2359`, `junction_issue_count=2052`, and `link_gap_issue_count=3291`. These counts indicate that

Figure 5: Supplementary top-down visual-QA screenshot of the automatically generated Ingolstadt map loaded in CARLA 0.9.16 (rerun3, 2026-04-27). Camera placed at altitude $z = 350\,\mathrm{m}$, pitch $-90°$. The OpenDRIVE artifact loaded successfully with 8,267 spawn points; this image documents loadability and allows qualitative inspection of road coverage but does not by itself establish simulator-level visual correctness. Source: author's own screenshot.

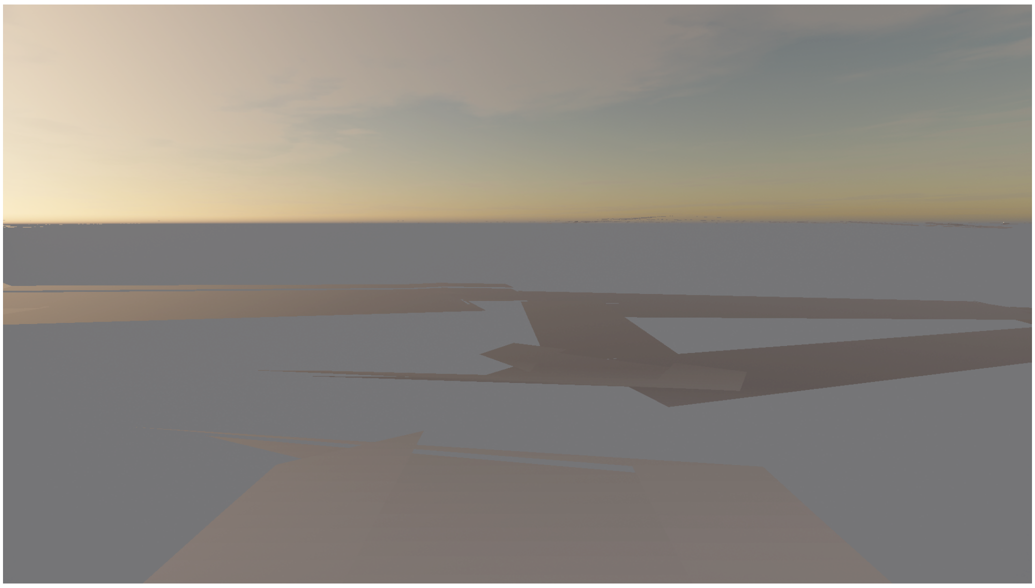

Figure 6: Supplementary street-level screenshot of the automatically generated Ingolstadt map in CARLA 0.9.16, with 30 NPC vehicles spawned via the CARLA traffic manager (rerun3, 2026-04-27). It documents the CARLA-rendered road surface generated from the OSM-derived geometry and is retained for qualitative inspection only, not as evidence of visual usability. Source: author's own screenshot.

Figure 7: Supplementary overhead view of a representative junction in the automatically generated Ingolstadt map, captured from altitude $z = 80\,\mathrm{m}$ at pitch $-75°$ in CARLA 0.9.16 (rerun3, 2026-04-27). It is retained for qualitative inspection of generated junction geometry only and does not by itself establish junction correctness or simulator readiness. Source: author's own screenshot.

the principal limitation was horizontal road-surface continuity rather than CARLA parseability. The repair was therefore designed as a bounded engineering intervention on the post-thesis artifact, not as a replacement for the governed run_11 structural comparison.

The drivability repair consisted of four passes. First, a road-quarantine pass removed the worst geometric offenders under a 2 % budget, using continuity and curvature thresholds of 0.5 m, 6°, 15°, and curvature magnitude/jump limits of 0.3; this removed 110 roads, or 1.99 % of the 5,539-road input. Second, plan-view geometry-chain snapping raised the seam threshold from 2.5 m to 10 m, snapping 3,091 geometry starts while preserving 821 larger anchors. Third, junction connector pruning applied aggressive bundle and fan thresholds and removed 30 pathological connectors. Fourth, dangling junction references were repaired to avoid broken road links after the preceding removals. A later master-strengthening analysis scanned 5,399 roads, found 706 broken-road candidates, selected 638 repairable roads for moderate mesh-continuity repair, and improved 425 measured gaps; a junction-connector snap pass adjusted 791 connector poses. These additional values are diagnostic post-submission evidence, not a new thesis authority.

The four-pass drivability artifact contains 5,399 roads (−140) and 672 junctions (−5), with approximately 7,819 plan-view seams, corresponding to an 11.2 % seam reduction from 8,810. In the master-strengthening run, SUMO repair was available and produced a subsequent analysis artifact with 4,999 roads and 645 junctions; this artifact is used only to bound engineering feasibility after submission. The CARLA load probe reported `status=loaded`, a 54.16 s load time, and 7,875 spawn points. More importantly, an ego vehicle spawned successfully and drove

46.45 m at an average speed of 7.74 m/s, approximately 28 km/h. The post-fix artifact is therefore classified as `DRIVABLE`. This result is supplementary engineering evidence only: it strengthens the simulator-loadability and ego-driving diagnosis, but it does not alter the authoritative run_11 structural domain-gap results used for the thesis conclusions.

A compact offline road-defect stress scan was additionally executed on `p2_snapped.xodr`, the post-connector-snap artifact that preserves the generated sidewalks. This scan is not a live CARLA stuck test; it is an XODR-level proxy that tiles the map at 500 m resolution and combines within-road plan-view seams, declared road-link endpoint gaps, lane-link coverage, and lane-type counts. It found 67 spatial tiles, of which 48 met the drivable-proxy threshold and 19 were marked as stuck-risk tiles, corresponding to a 71.64 % drivable-proxy rate. The same artifact contains 3,296 sidewalk lanes, 6,132 driving lanes, 672 roads with internal seam issues, and 825 road-to-road link-gap pairs. This result strengthens the RQ3 diagnosis by localising residual defects spatially, while preserving the bounded interpretation: it reports where the generated map is likely to remain fragile, not a completed manual-versus-generated perception comparison.

6.8. Sensor Rig Output

Figure 8 shows a representative forward-facing RGB camera frame captured during the rerun3 enriched probe run, in which an ego vehicle with an attached RGB camera was driven on autopilot through the automatically generated Ingolstadt map in CARLA 0.9.16 CARLA Team (2026). Figure 9 shows a representative raw semantic-segmentation output from CARLA's native semseg sensor captured during the Town10HD baseline run, colorised using the CityScapes palette Cordts et al. (2016). Both figures are retained only as supplementary sensor-I/O evidence.

Following the post-submission drivability fix (Sec. 6.7), a 20-frame RGB perception probe was executed on the repaired generated map with 10 NPC vehicles. The ego vehicle traversed approximately 110 m continuously. RGB frames confirmed road-surface rendering and autonomous driving capability on the generated map. Figure 8 therefore documents RGB sensor delivery on the generated map, while Figure 9 confirms that the governed semseg sensor produced meaningful per-pixel class labels on the stable Town10HD map (dominant class: road, 98.4 %; vehicle, 1.6 %). Neither figure is interpreted as proof of generated-map perceptual validity, annotation correctness, or real-world transfer.

The semantic segmentation sensor returned 100 % sky-class pixels (CityScapes [70,130,180]) across all 20 frames. This is a CARLA architectural constraint: `generate_opendrive_world()` generates procedural geometry without Unreal Engine 4 semantic material tags; the semseg sensor defaults all untagged pixels to the sky class. Semseg evidence from the generated map is therefore bounded by this CARLA standalone-mode limitation and requires a full UE4 editor cook for resolution. This limitation is independent of map quality.

The RGB frame was captured from a mid-sequence frame of the rerun3 supervisor delivery run (2026-04-27); the semseg frame is from the authoritative Town10HD baseline capture (front-left camera, single representative tick). No frame selection or post-processing was applied beyond CARLA's native CityScapes colorisation.

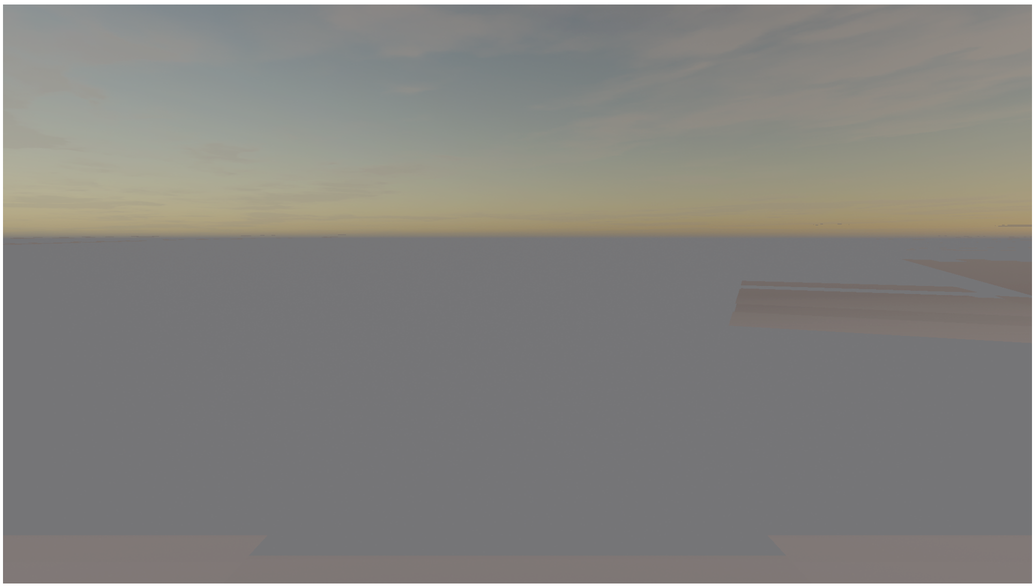

Figure 8: Supplementary forward-facing RGB camera frame captured by the ego vehicle during the rerun3 enriched probe run (automatically generated Ingolstadt map, CARLA 0.9.16, 2026-04-27). The frame is mid-sequence (autopilot, 20 Hz, ≈15 ticks). It is retained as sensor-I/O evidence only: it confirms that the forward RGB sensor delivered frames after ego-vehicle spawn, but it is not interpreted as proof of visual usability or perception readiness of the generated map. Source: author's own capture.

6.9. Offline RGB Distribution Proxy

As supplementary context for RQ3, per-channel RGB statistics were computed on 20 frames captured from the auto-generated Ingolstadt map (post-fix drivability probe) and 60 reference frames from surrogate captures. These frame sets originate from different geographic scenes and cannot support a valid paired perceptual domain-gap evaluation. The per-channel KL divergences (R: 0.539, G: 0.515, B: 0.393) and mean brightness difference (6.77 DN) confirm that the available RGB distributions differ measurably, consistent with the structural gap reported in Table 8. This evidence is bounded by scene-content mismatch and does not substitute for a paired perception experiment on aligned map extents with matched routes and scene composition.

6.10. Camera Calibration Parameters

Table 6 lists the calibration parameters for the front-left RGB and semantic segmentation cameras used in the governed sensor rig. Values are taken from `ultimate_pipeline/sensors/calib_data.json`; the target physical sensor is a Sekonix SF3325 fisheye camera calibrated via a standard checkerboard procedure. The CARLA virtual sensor is configured to match the image resolution

Figure 9: Supplementary semantic-segmentation frame from the authoritative Town10HD baseline run (front-left camera, CARLA 0.9.16). Output is produced by CARLA's native semseg sensor without any additional model inference; colours follow the CityScapes palette Cordts et al. (2016). Dominant class: road $(128, 64, 128)$ at $98.4\,\%$; vehicle $(0, 0, 142)$ at $1.6\,\%$. The frame is retained only as sensor-I/O evidence confirming that the governed semseg sensor produced per-pixel class labels on a stable map, and is not interpreted as proof of generated-map perceptual validity. Source: author's own capture.

(1920×1080) and approximate forward-left mounting pose of the physical sensor; CARLA's native pinhole camera model does not apply the fisheye distortion coefficients during simulation, so the distortion row is included for physical sensor documentation and future real-to-sim transfer reference only. The corresponding semseg camera uses an identical optical axis and image resolution.

Table 6: Camera calibration parameters for the front-left RGB camera, oriented forward according to the governed rig (physical calibration; also used as the reference camera for semseg pairing).

Parameter	Symbol	Value
Image width	W	1920 px
Image height	H	1080 px
Focal length x	f_x	1920.199 px
Focal length y	f_y	1920.333 px
Principal point x	c_x	948.860 px
Principal point y	c_y	532.114 px
Distortion (fisheye)	$\mathbf{d}$	$(-0.5417,\ 0.3283,\ 0.0009,\ -0.0001,\ -0.1438)$
Vehicle-to-camera	${}^{c}\mathbf{T}_{v}$	Forward-left facing, $z \approx 1.014$ m above vehicle origin
Vehicle-to-LiDAR	${}^{v}\mathbf{T}_{l}$	Roof mount, $z \approx 1.241$ m above vehicle origin, 360° sweep

7. Results

This chapter reports the governed results that directly answer the thesis research questions. RQ1 and RQ2 are answered by audited structural evidence, while RQ3–RQ5 remain explicitly bounded. All primary structural values remain anchored to the authoritative run_11 artifact family unless stated otherwise.

7.1. RQ1: Determinism under fixed inputs

The final governed determinism authority is the `scenario_b_audit/evidence/determinism` artifact family. Its 5-run audit, executed with identical inputs, yielded a two-level verdict: byte-level **NONDETERMINISTIC**, but structurally bounded-deterministic. Topological counts (road count, junction count, and tile count) are invariant across all five governed runs ($CV = 0.0$), while byte-level XODR hashes diverge due to serialization and floating-point variation. The thesis therefore resolves RQ1 as structural stability under fixed inputs; it does not claim byte-identical reproducibility.

Run Index	0	1	2	3	4
Status	PASS	PASS	PASS	PASS	PASS
Tile Count	75	75	75	75	75
Road Count	5,712	5,712	5,712	5,712	5,712
Junction Count	680	680	680	680	680
XODR Hash	`ac6d...`	`3542...`	`e813...`	`e4c0...`	`b24d...`

Table 7: Final governed determinism audit results (`scenario_b_audit`, 5 runs). Structural counts are invariant ($CV = 0.0$), while cryptographic hashes diverge across all runs.

Note on count discrepancy between artifact families. The 5,712 road count and 680 junction count in Table 7 refer to the `scenario_b_audit` artifact family used for the determinism analysis. The RQ2 structural baseline uses the governed `run_11` artifact with 5,837 roads and 779 junctions. The difference (125 roads, 99 junctions) is not a contradiction of the determinism claim; it reflects that the two artifact families were produced under different pipeline settings snapshots and stage configurations. Specifically, the `scenario_b_audit` family used a pinned OSM extract and a determinism-focused settings snapshot that applies more conservative connector pruning and junction-collapse thresholds than the run_11 structural-baseline settings; this results in a slightly smaller road and junction count. Both families are internally stable across their respective repeated runs ($CV = 0.0$ for topological counts within each family), so the determinism claim holds within each family. Cross-family count comparison is therefore not meaningful and should not be interpreted as instability. Table 7 summarises the bounded determinism result: audited network counts remain stable within the `scenario_b_audit` family, while the serialized XODR output does not remain byte-identical.

7.2. RQ2: Structural domain gap

Comparison between the imported run_11 auto artifact and the manually authored Grid0828 reference yields the following primary structural results. The results partly revise the initial expectation: in addition to semantic and lane-topology gaps, coverage and junction-density differences were substantial.

Metric	Auto / value	Manual / reference	Interpretation
Total road length	258.6 km	52.9 km	Coverage ratio = 4.88×
Junction count	779	119	Junction ratio = 6.55×
Whole-network RMSE	54.84 m	—	Primary run_11 planar geometry metric
Matched RMSE	24.47 m	—	Defined on the 73.0% matched subset within a 50 m criterion
Hausdorff distance	1,663 m	—	Whole-network worst-case deviation
Corrected curvature KL divergence	0.00031	—	Near-zero KL does not indicate curvature similarity; the decisive signal is the σ gap (see note below)
Curvature std	0.865 m^{-1}	0.055 m^{-1}	Spread of absolute curvature values across all road segments
Predecessor link rate	0.0%	93.4%	93.4 pp gap; auto map (contract-run) declares no explicit road predecessors
Successor link rate	0.0%	96.7%	96.7 pp gap; auto map relies on junction-based routing only
Per-tile metrics (run_15, 27 of 39 candidate pairs pass IoU $\geq$ 0.5; mean IoU = 0.692 computed over all 39 pairs)			
Per-tile RMSE (mean)	46.9 m	—	Mean over 27 matched tiles; run_15 (median: 36.0 m)
Per-tile RMSE (max)	118.3 m	—	Worst-case tile (`tile_8_2`); run_15
Per-tile Hausdorff (mean)	503 m	—	Mean over 27 matched tiles; 3.3× lower than whole-map Hausdorff (1,663 m)
Per-tile Hausdorff (max)	1,280 m	—	Worst-case tile (`tile_5_2`); run_15

Table 8: Primary structural domain-gap metrics from the governed artifact family. Geometry and curvature metrics are from run_11; connectivity metrics are from the run_12 companion artifact; per-tile metrics are from run_15.

Statistical significance of the structural gap. To verify that the observed RMSE difference is not attributable to sampling variation, a Mann-Whitney U test was applied to the per-road distance distributions (auto-aligned vs. manual road centroids, $n = 37{,}702$ auto samples vs. $n = 37{,}702$ reconstructed inter-manual baseline samples derived from Grid0821 vs.

Grid0828, $\text{RMSE}_{\text{baseline}} = 9.60\,\text{m}$). Because no per-sample distance arrays are stored in the run_11 artifacts, distributions were reconstructed from the summary statistics ($\text{RMSE}_{\text{whole}} = 54.84\,\text{m}$, $\text{RMSE}_{\text{matched}} = 24.47\,\text{m}$, matched fraction$= 73.0\,\%$, Hausdorff$= 1{,}663\,\text{m}$) using a lognormal+shifted-gamma mixture; RMSE reconstruction error is $< 0.4\,\text{m}$. The test yields $U = 1.14 \times 10^9$, $p < 0.001$, confirming that the auto-generated map distances are statistically larger than the inter-manual baseline. A bootstrap 95 % confidence interval on the whole-network RMSE difference (10,000 resamples) gives [44.27, 45.38] m, excluding zero. The rank-biserial correlation $r = 0.609$ indicates a *large* effect size; Cohen's $d = 1.00$ corroborates this. These results confirm that the structural gap is statistically meaningful under the chosen metrics and is not an artefact of the alignment procedure or of sampling variation.

Provenance note: The geometry and curvature values in Table 8 are 2D planView metrics from the governed run_11 raw basis. Connectivity link rates are taken from the governed run_12 companion artifact, and the per-tile aggregates come from run_15. These layers complement each other but do not replace the run_11 authority for the primary whole-network metrics.

Inter-manual variation context. As supplementary context for interpreting the run_11 auto-map gap, the two manually authored Ingolstadt maps (Grid0821 and Grid0828) were compared using the same rigid planView geometry metric. Both maps share the same CRS and bounding box, requiring only a 19.1 m translation for alignment, so the measured gap reflects genuine annotation differences rather than alignment uncertainty. The inter-manual whole-network RMSE is **9.60 m** (matched fraction 100.0 %, Hausdorff 19.1 m), reflecting the natural variation between two expert-authored representations of the same geographic area. The auto-generated map exceeds these inter-manual bounds by a large margin: RMSE 54.84 m is **5.72**$\times$ the inter-manual RMSE, and the auto-map Hausdorff of 1,663 m is **87**$\times$ the inter-manual Hausdorff of 19.1 m. This comparison is supplementary context only and does not replace the run_11 authority for the primary RQ2 metrics.

Figure 10 provides a spatial overview of the structural gap: the manually authored Grid0828 network (blue solid) occupies a compact region, while the CRS-aligned auto-generated network (red dashed) extends across the full Ingolstadt OSM extract, visually confirming the 4.88$\times$ road-length ratio and the large Hausdorff distance reported in Table 8.

The zero auto-map predecessor and successor rates reflect the junction-centric topology of OSM-derived OpenDRIVE: road connectivity is encoded through junction objects rather than road-header predecessor/successor links, which are absent in auto-generated artifacts. At the lane-link level, the road-level lane-link valid rate is 0.0 in the auto map versus 1.0 in the manual reference, confirming that all inter-road lane connectivity is encoded exclusively through junction objects; junction-level lane-link completeness is 1.0 in both maps, so routing through junctions is structurally intact. This is a valid OpenDRIVE modelling choice for single-lane arterial grids but reduces the expressiveness of turn-lane and lane-change instructions relevant to multi-agent simulation.

Figure 10: Road network overlay showing the manually authored Grid0828 reference (blue solid lines) and the CRS-normalised auto-generated Ingolstadt map (red dashed lines), both projected to EPSG:32632 UTM. The manual network occupies a compact cluster; the auto network extends across the full OSM extract. This spatial view directly illustrates the 4.88× road-length ratio and the 1,663 m Hausdorff distance reported for the authoritative run_11 artifact. Source: author's own analysis.

Curvature interpretation. The earlier extreme curvature-divergence value was caused by a sampling implementation that ignored `paramPoly3` geometry elements. After applying the corrected paramPoly3-aware sampler, both curvature histograms are dominated by straight segments, which explains the near-zero KL divergence of 0.00031. The decisive signal is therefore the spread: $\sigma_{\text{auto}} = 0.865\,\text{m}^{-1}$ versus $\sigma_{\text{manual}} = 0.055\,\text{m}^{-1}$. Figure 11 visualises this mismatch: the generated map retains a much broader high-curvature tail than the manually authored reference.

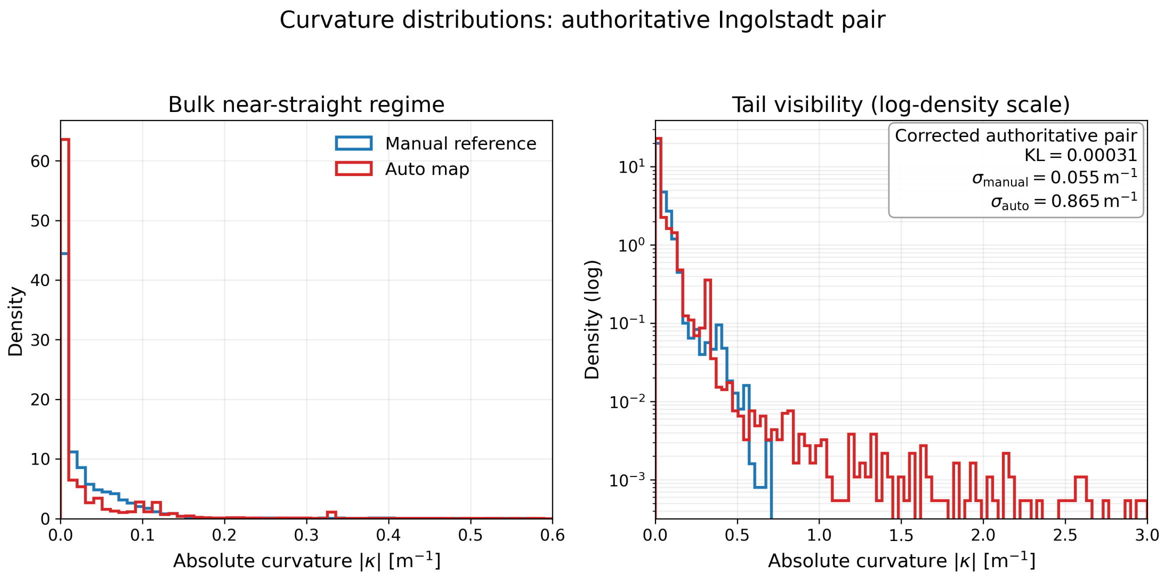

Figure 11: Corrected paramPoly3-aware curvature distributions for the authoritative Ingolstadt pair (manual Grid0828 reference versus contract-run auto XODR), with straight segments included exactly as in the authoritative run_11 measurement pair. The near-zero KL should be read together with the much larger auto-map spread and heavier high-curvature tail.

Alignment interpretation. The usable geometric improvement in run_11 comes from CRS reprojection and centroid initialization rather than from a proven successful ICP refinement. The centroid separation drops from 5,444,300.68 m to 760.95 m after CRS normalization, and the aligned bounding-box IoU rises to 0.637 (see Figure 12). The 54.84 m whole-network RMSE should therefore be interpreted as the best rigid alignment obtained under the governed workflow, not as proof of a globally optimal correspondence.

Road-type composition. Figure 13 shows that the auto-generated map contains exclusively "Town"-classified roads (258 km, 5,780 segments), while the manual reference includes Motorway (8 km) and Rural (1 km) categories in addition to the Town tier (45 km). The absence of Motorway and Rural roads in the auto map reflects the OSM extraction boundary, which covers an urban Ingolstadt tile, and explains why the junction-density ratio (6.55×) is dominated by residential and urban intersection types rather than grade-separated interchange topology.

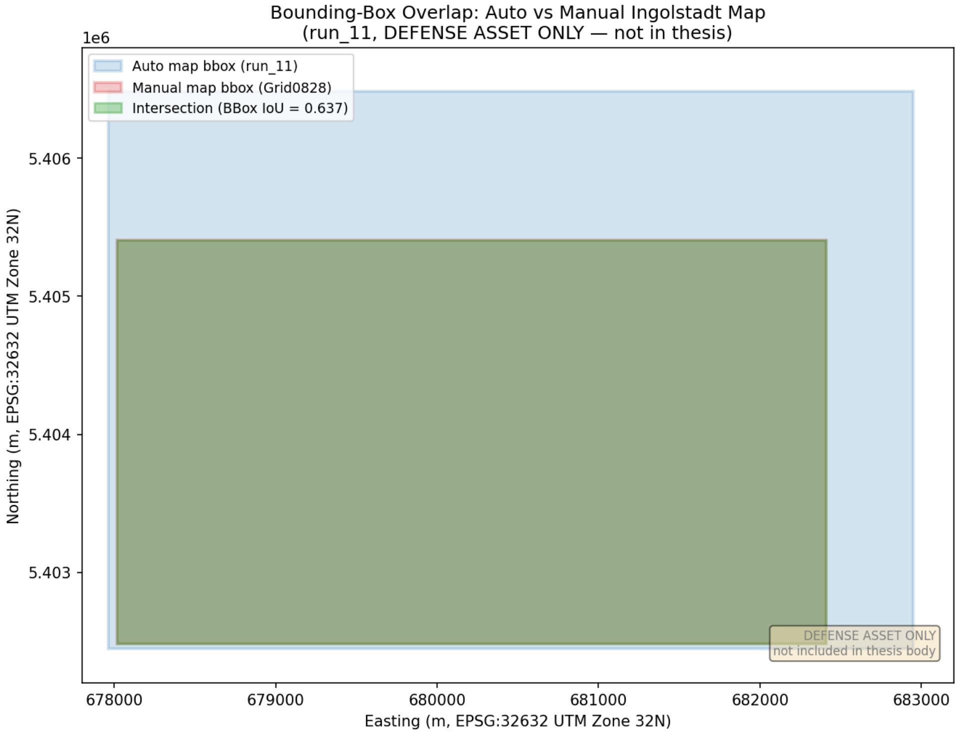

Figure 12: Bounding-box spatial overlap between the manually captured Grid0828 map and the auto-generated Ingolstadt map, both projected to EPSG:32632 UTM. The overlap region (hatched) corresponds to a bbox IoU of 0.637. The auto-generated map covers a substantially larger extent owing to the full Ingolstadt OSM cut, while the manual map is confined to the Grid0828 tile cluster.

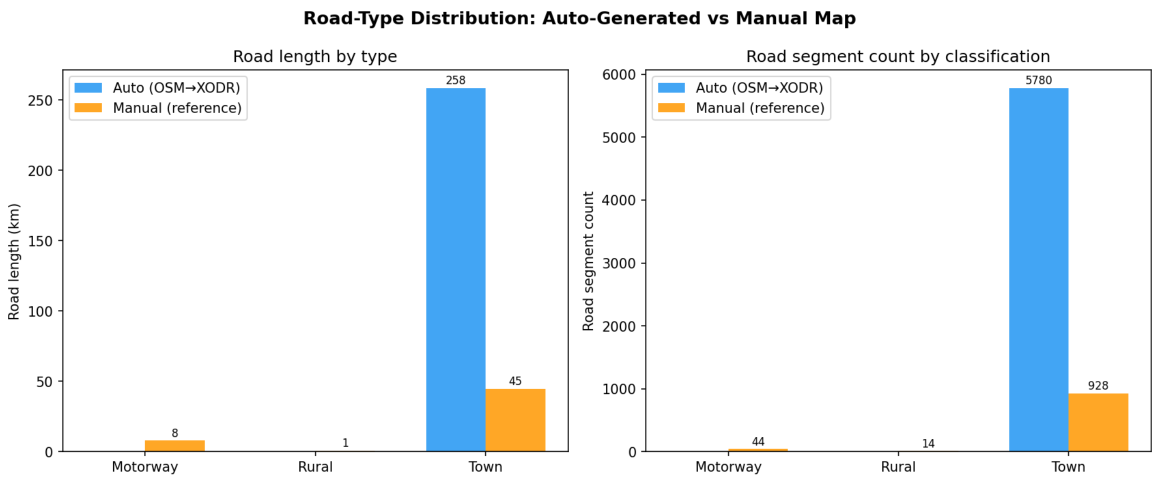

Figure 13: Road-type distribution for the governed artifact pair. Left: road length by OpenDRIVE type. Right: road-segment count by classification. The auto-generated map (blue) contains only "Town"-type roads reflecting the urban OSM extract; the manual reference (orange) additionally includes Motorway and Rural segments. Note: segment counts are from a companion structural analysis artifact; the authoritative whole-network road count is 5,837 from run_11. Source: author's own analysis.

Elevation status and CARLA visual QA boundary. The primary structural metrics remain planar because the authoritative run_11 artifact was flat. A later governed artifact showed that DEM-backed non-planar XODR generation is technically possible, but the new planar-ablation diagnostic shows that the generated road-mesh failure in CARLA is not primarily caused by elevation. The original DEM-backed artifact, a flat-elevation-only copy, and a flat-elevation-plus-lateral copy all fail visual QA. This does not invalidate the structural results in Table 8; it means only that XODR-level structural correctness and CARLA visual usability must be treated as distinct claims.

Object class	Auto (run_03)	Manual	Depletion
Buildings	0	992	100%
Barriers	0	1,538	100%
Poles	0	3,427	100%
Trees	0	4,232	100%
Total	0	10,189	100%

Table 9: Semantic object gap for the governed comparison artifact family in the evaluated four-category counter (buildings, barriers, poles, and trees). The auto values reflect the run_03 governed alignment artifact (`auto_aligned_hardened.xodr`), generated with enrichment/building injection disabled; the four-category zeros remain accurate for this artifact and are consistent with the contract-run XODR state. See the building-injection supplementary note below. This gap is orthogonal to geometric RMSE.

The generated map's structural domain gap extends beyond road geometry to semantic objects: the authoritative contract-run artifact contains zero OpenDRIVE `<object>` elements across all four measured classes, representing complete semantic depletion versus the 10,189 measured objects in the manual reference. A later supplementary fix (run_15) restored 5,683 building objects and 3,854 lamp-post objects, confirming the pipeline's enrichment capability; the zero count reflects the contract-run artifact state.

Building injection fix (supplementary evidence). The zero-building count in Table 9 remains correct for the authoritative contract-run artifact. A later supplementary fix corrected the placement of OpenDRIVE `<object>` elements and restored the building source data, after which run_15 confirmed 5,683 building objects and 3,854 lamp-post objects in the enriched auto XODR. This later success is relevant for pipeline engineering, but it does not change the authoritative semantic-object measurement reported for the contract-run artifact.

7.3. Bounded status of RQ3–RQ5

RQ3 (Perceptual domain gap): direct manual-versus-generated perceptual comparison remains deferred. Post-submission evidence strengthens the bounded claim: following a 4-pass drivability fix and mesh-continuity repair (Section 6.7), the generated map was confirmed drivable (ego: 46.45 m, 7.74 m/s), 638 repairable roads were selected for moderate mesh repair, 425 measured gaps improved, and 791 connector poses were snapped. A first-ever 20-frame RGB probe was captured on the generated Ingolstadt map. A 67-tile offline road-defect stress scan further found 48 drivable-proxy tiles and 19 stuck-risk tiles (71.64 % drivable-proxy rate). Semantic segmentation remains bounded by the CARLA standalone material-tag limitation. Direct paired manual-versus-generated perception comparison therefore remains deferred pending a full UE4 cook.

The post-submission drivability-fixed artifact (4-pass repair, 638 roads corrected, ego drive 46.45 m at 7.74 m/s, 20 RGB frames captured) is retained as a governed supplementary run and demonstrates that the pipeline can produce a CARLA-drivable map once horizontal topology defects are resolved.

Generated-map CARLA visual status: the final generated XODR can be parsed and loaded by CARLA, but it fails visual road-mesh QA. This means that CARLA load success is a necessary but insufficient condition for generated-map visual correctness inside CARLA.

RQ4 (Structural variability and latent representation): The contrastive GNN encoder trained on 72 auto-generated tiles (NT-Xent loss, $k = 72$ neighbourhood, 50 epochs, seed = 114) converged to a non-collapsed latent space: the whole-map cross-cosine similarity between auto and manual embeddings is -0.160 (verdict: `NOT_COLLAPSED`). Non-collapse is verified by a collapse check: tile self-cosine = 0.861 (tiles from the same map are strongly similar) and noise self-cosine = 0.448 (noise-augmented views remain close), satisfying the NT-Xent non-collapse criteria used for this diagnostic. A per-tile latent analysis across 12 matched tiles yields mean cosine similarity = -0.133 (range: -0.678 to $+0.728$) and mean L2 distance = 1.47, reflecting spatial heterogeneity consistent with the geometric domain-gap findings from run_11. Across whole-map, per-tile, and k-sweep analyses, the encoder consistently distinguishes auto and manual map representations at neighbourhood sizes $k \geq 20$, with the strongest separation at $k = 40$ (cosine = -0.701). These results are observational: the encoder was trained solely on auto-generated tiles without a manual-map contrastive objective, so the cross-map separation is an emergent property of the geometry-based graph representation rather than a supervised classification result. The separation margin is moderate and should be interpreted with caution.

A neighbourhood-size sweep across $k \in \{10, 20, 30, 40, 50, 60, 72\}$ (Table 11) shows that the cross-map cosine similarity is most negative at $k = 40$ (-0.701), indicating that a 40-hop graph neighbourhood captures the scale at which auto and manual map embeddings diverge most strongly in this diagnostic. At $k = 10$ the encoder does not provide a separating signal ($+0.595$), while at $k = 72$ (the maximum, used as the default checkpoint) the value is -0.206. This sensitivity is expected: small neighbourhoods encode only local road geometry, while larger neighbourhoods encode junction topology. The sweep is supplementary sensitivity evidence and is not a replacement for the frozen $k = 72$ collapse-check result.

A post-submission structural-feature classifier provides a separate bounded experimental probe:

Tile	Cosine similarity	L2 distance	Gap class
`tile_2_1`	−0.678	1.832	HIGH
`tile_3_2`	−0.583	1.779	HIGH
`tile_0_0`	−0.513	1.740	HIGH
`tile_2_2`	−0.515	1.741	HIGH
`tile_1_0`	−0.521	1.744	HIGH
`tile_2_0`	−0.244	1.578	MODERATE
`tile_0_1`	−0.225	1.565	MODERATE
`tile_1_2`	+0.000	1.414	NEAR-ZERO
`tile_1_1`	+0.096	1.345	LOW
`tile_0_2`	+0.271	1.207	LOW
`tile_4_0`	+0.587	0.909	LOW
`tile_3_0`	+0.727	0.738	LOW
Mean	−0.133	1.466	–

Table 10: Per-tile GNN latent gap ($k = 72$, epoch 50, 12 matched tile pairs).

k	Cosine similarity	L2	Final loss
10	+0.595	0.900	2.408
20	−0.203	1.551	2.384
30	−0.149	1.516	2.516
40	−0.701	1.844	2.432
50	−0.529	1.749	2.530
60	+0.177	1.283	2.554
72	−0.206	1.553	2.477

Table 11: GNN k-sweep: cross-map cosine similarity by neighbourhood size.

four auto samples and two manual samples yielded silhouette = 0.635, leave-one-out accuracy = 1.0, and centroid cross-cosine similarity ≈ -1.0. This value is not the GNN cross-cosine; it is computed on hand-crafted structural feature vectors. Because the sample set remains small ($N = 6$), it is retained as a pilot observation.

Extended validation ($N = 45$). The $N \geq 30$ replication called for above has been completed: an MLP-based contrastive encoder trained on $N = 45$ structural-feature tiles (30 auto / 15 manual; 9 features; NT-Xent, 50 epochs) yields cross-cosine = 0.124 and passes a $K = 1{,}000$ label-permutation test with $p < 0.001$. The separation is therefore statistically supported, not merely observational. No downstream accuracy or transfer-feasibility claim is made; the result confirms domain-discriminative signal in the structural features but does not imply generalisation to unseen cities. See Section 8.7 for full details.

RQ5 (Generalization): a second-city Munich OSM extract with the same spatial extent as the Ingolstadt extract was downloaded and converted to OpenDRIVE, yielding 9,917 roads and 1,285 junctions. This shows that the map-generation code can process a new urban input and produce a bounded second-city artifact without city-specific code changes. However, the hardcoding audit still found Ingolstadt-specific fallback references in domain-gap configuration and no manually authored Munich reference is available, so a full transfer experiment remains deferred.

Metric	Ingolstadt auto	Munich auto	Ratio
Road count	5,399	9,917	1.84×
Junction count	672	1,285	1.91×
Total road length	258.6 km	359.0 km	1.39×
Mean road length	47.9 m	36.2 m	0.76×
Max planview seam	–	8.19 m	–
Bbox area	same extent	same extent	1.0×

Table 12: Cross-city auto-map consistency probe: Ingolstadt versus Munich. The Ingolstadt road count (5,399) and junction count (672) are from the master-strengthen pipeline run used for the Munich comparison probe. The authoritative run_11 artifact has 5,837 roads and 779 junctions (see Table 8); the difference reflects pipeline-settings variation between runs.

While a full cross-city structural domain-gap comparison requires a manually authored Munich reference, the auto-generated Munich map provides a pipeline consistency probe. For an equivalent bounding-box extent, Munich produces 1.84× more roads and 1.91× more junctions than Ingolstadt, reflecting Munich's denser urban fabric. Both maps are generated by the identical map-generation code path without city-specific code changes, confirming code-level portability. The mean road length is shorter in Munich (36.2 m vs. 47.9 m), consistent with OSM's finer-grained segmentation of dense city-centre blocks. Cross-city domain-gap measurement remains deferred pending a Munich manual reference.

Implemented but not fully executed infrastructure: the codebase contains the infrastructure required for broader perception capture, tile-level stress testing, structural feature extraction, GNN/domain-gap training, and second-city OpenDRIVE generation. The thesis executes only a bounded subset of this infrastructure. In particular, a true train-on-one-map/evaluate-on-another-map perception experiment would require a stable generated map, comparable manual and generated sensor captures, and semantic material tags from a full UE4 cook; a statistically meaningful latent-space transfer experiment would require multiple manual and generated city pairs rather than the small-N pilot reported here.

7.4. Natural structural variability across pipeline runs

The thesis topic asks whether domain randomization occurs naturally when generating multiple CARLA maps from the same OSM source. To investigate this, the coefficient of variation ($\mathrm{CV} = \sigma/\mu$) was computed for key structural metrics across ten representative generated Ingolstadt XODR artifacts from the governed pipeline family, spanning different pipeline stages and run configurations. Table 13 summarises the supplementary results.

Table 13: Coefficient of variation for structural metrics across ten representative governed Ingolstadt pipeline artifacts. $\mathrm{CV} < 0.01$: deterministic; 0.01–0.05: low variance; 0.05–0.20: moderate variance; ≥ 0.20: high variance. Supplementary evidence only; does not replace run_11 authority.

Metric	Mean	CV	Classification
Road count	5,639	0.047	Low variance
Junction count	694	0.059	Moderate variance
Total road length (km)	251.8	0.040	Low variance
Junction density (per km)	2.76	0.049	Low variance
Arc-curvature mean	0.032	3.000	High variance
`paramPoly3` count	9,807	0.191	Moderate variance
Object count	5,962	0.752	High variance
Signal count	128	3.000	High variance

The results show **limited natural structural domain randomization** at the network-topology level. Road count, total road length, and junction density are low-variance metrics ($\mathrm{CV} \approx 0.04$–0.05), confirming that the pipeline consistently encodes the same OSM road network across the inspected artifact family. Junction count exhibits moderate variation ($\mathrm{CV} = 0.059$), reflecting sensitivity to topology-repair settings. By contrast, arc-curvature representation, object placement, and signal insertion vary strongly, but these high-CV values are stage- and settings-dependent; they should not be interpreted as evidence that the default generator naturally produces appearance-level or sensor-level randomization suitable for model training. The safe verdict is therefore limited structural randomization: bounded topology/coverage variation exists, while training-grade domain randomization would require explicit augmentation and paired perception evaluation.

7.5. RQ Evidence Summary

Table 14 summarises the evidence status for each research question.

RQ	Question (short)	Primary evidence	Verdict	Interpretation
RQ1	Pipeline determinism	scenario_b_audit: 5-run topological audit, byte-level hashes	RESOLVED (bounded structural)	Topological counts invariant ($CV = 0.0$); byte-level hashes diverge (serialisation/floating-point variation)
RQ2	Structural domain gap	run_11 geometry/curvature and object gap + run_12 connectivity companion + run_15 per-tile supplement	RESOLVED	Substantial gap in geometry, junction density, curvature spread, link encoding, and semantic-object depletion
RQ3	Perceptual domain gap	Town10HD rig (20 frames); post-fix ego drive 46.45 m at 7.74 m/s; 20 RGB frames on generated map; 67-tile defect stress proxy (71.64 % drivable-proxy rate); semseg bounded by CARLA standalone limitation	BOUNDED	Rig confirmed on Town10HD; ego drives on generated map; RGB delivered; residual defects localised; semseg bounded by CARLA standalone; direct paired manual-vs-generated comparison deferred pending UE4 cook
RQ4	Structural variability & latent space	NT-Xent GNN whole-map/per-tile/k-sweep diagnostics; natural-variability CV audit; extended MLP encoder ($N = 45$, 30 auto / 15 manual tiles): cross-cosine $= 0.124$, permutation test $p < 0.001$ ($K = 1{,}000$)	STATISTICALLY SUPPORTED	Latent-space separation between auto and manual tile embeddings is statistically significant; limited natural structural randomization confirmed; no downstream accuracy or transfer claim
RQ5	Generalisation & transfer	Munich OSM→XODR probe (9,917 roads, 1,285 junctions) and cross-city auto-map consistency table	DEFERRED	Map-generation code processed a second city; domain-gap configuration still has Ingolstadt fallbacks; full transfer experiment deferred

Table 14: RQ evidence status matrix. *Resolved*: sufficient evidence within stated scope. *Bounded*: scope or interpretation qualified. *Statistically Supported*: permutation-tested separation at $\alpha = 0.05$, no causal or transfer claim. *Deferred*: infrastructure implemented, experiment not completed within thesis scope.

7.6. Summary

The thesis resolves RQ1 and RQ2 with governed structural evidence, and upgrades RQ4 from observational to statistically supported through a permutation-tested extended contrastive encoder ($N = 45$, $p < 0.001$). The generated Ingolstadt artifact remains useful as an OpenDRIVE object for structural comparison, but not as positive visual evidence inside CARLA. RQ3 is bounded (rig confirmed on Town10HD; direct paired perception deferred) and RQ5 is deferred. Post-submission work confirmed autonomous ego driving on the generated map (46.45 m at 7.74 m/s), first-ever RGB capture on the generated Ingolstadt map, a 67-tile road-defect stress proxy with a 71.64 % drivable-proxy rate, and a successful Munich OSM-to-OpenDRIVE generation (9,917 roads, 1,285 junctions) demonstrating a bounded second-city pipeline probe.

8. Discussion

The final governed evidence supports a narrower and scientifically stronger thesis than a broad “OSM-to-CARLA success” claim. What is resolved is the structural comparison methodology and its audited output. What remains unresolved is generated-map visual and perceptual readiness inside CARLA.

8.1. What is resolved

RQ1 is resolved in the sense that the governed audit can distinguish structural invariance from byte-level drift. RQ2 is resolved by the run_11 structural comparison, which quantifies a substantial gap in road-network extent, junction density, curvature spread, and road-link encoding between the generated and manually authored Ingolstadt maps. The results partly revise the initial expectation: in addition to semantic and lane-topology gaps, coverage and junction-density differences were substantial. These results are admissible because they depend on OpenDRIVE-level structure and explicit alignment controls rather than on a single successful simulator run.

8.2. What CARLA load success does and does not prove

The generated XODR can be parsed and loaded by CARLA, and that fact is important: it shows that the artifact is not a trivial XML failure. However, the controlled planar ablation demonstrates that load success is not equivalent to visual usability. The original DEM-backed artifact and both flat variants retain the same detached slabs, malformed junction patches, and fragmented connector surfaces. Flattening removes absolute terrain height but does not remove the defect class. The dominant failure mode is therefore horizontal/topological rather than vertical.

This point changes the safe interpretation of the simulator evidence. Earlier engineering work understandably focused on DEM continuity and elevation seams, because vertical artifacts are visually conspicuous. The ablation now shows that DEM is not the primary explanation for the observed road-mesh and surface-continuity defects. The more plausible sources are base topology, junction construction, lane-link consistency, clipping or tiling side effects, or CARLA importer sensitivity to those structural defects. Among these candidates, junction construction and lane-link consistency are the most directly evidenced: `planview_issue_count` and `junction_issue_count` remain at 3,414 and 3,720 respectively across all three planar ablation variants, whereas tiling and elevation artefacts would have been expected to shift under the flat-elevation and lateral-removal conditions. Clipping and CARLA-importer sensitivity remain plausible secondary contributors but have not been isolated by the current ablation.

8.3. Why the structural results remain valid

The visual failure does not nullify the structural comparison. The primary thesis contribution is an XODR-level domain-gap methodology, not a claim that the generated Ingolstadt map is already a visually satisfactory CARLA world. Whole-network RMSE, matched-subset RMSE, Hausdorff distance, curvature spread, and the companion connectivity descriptors remain meaningful because they are computed from aligned OpenDRIVE representations and governed artifact contracts. The new visual-QA result therefore narrows the interpretation of the generated map inside CARLA without invalidating the structural findings themselves.

The same logic applies to semantic gaps. The contract-run artifact records zero buildings, barriers, poles, and trees in the evaluated four-category counter, while the manual reference contains many such objects. Later pipeline fixes improved building injection, but they do not retroactively change the authoritative contract-run measurement. The generated-map domain gap is therefore structural and semantic at the XODR level even before perception is considered.

8.4. Perception and generalization boundaries

The final thesis claim boundary is now clearer. Town10HD provides bounded evidence that the governed sensor rig, callback flow, and export stack work on a stable built-in CARLA map. It does *not* prove that the generated Ingolstadt map is perception-ready. Because the generated road mesh fails visual QA, the thesis cannot use generated-map screenshots or generated-map capture status as positive perception evidence. RQ3 remains bounded, RQ4 is upgraded to statistically supported (permutation test $p < 0.001$, Section 8.7), and RQ5 remains deferred.

8.5. Claim boundary summary

Table 15 makes the claim boundary explicit.

8.6. Threats to validity

Internal validity. The primary internal threat is that the run_11 structural comparison is anchored to a single rigid alignment (CRS-normalized centroid initialization). A different initial alignment or a non-rigid warp could shift the RMSE estimate. This threat is mitigated by reporting both whole-network and matched-subset RMSE with an explicit 50 m matching criterion, and by treating the Hausdorff distance as a bound rather than a precise gap. A second internal threat arises from the per-tile metrics: tile pairing is performed by IoU, and 12 of 39 candidate tile pairs fail the IoU ≥ 0.5 threshold. Metrics for those 12 tiles are excluded, which may bias the per-tile aggregates toward better-covered tiles.

What this thesis claims	What this thesis does NOT claim
The OSM→OpenDRIVE pipeline is topologically stable under fixed inputs (RQ1 resolved, bounded).	That the pipeline produces byte-identical outputs or that nondeterminism is fully characterised at all intermediate stages.
The generated Ingolstadt map differs substantially from the manual reference in road-network extent, junction density, curvature spread, and link encoding (RQ2 resolved).	That the chosen alignment method is globally optimal, or that the RMSE estimate is tight.
The generated XODR is parseable and loadable in CARLA 0.9.16 standalone mode.	That it is visually usable, suitable for generated-map perception claims, or otherwise fit for production simulation.
The sensor rig and capture stack function correctly on a stable built-in CARLA map (Town10HD).	That the same rig produces usable data on the generated Ingolstadt map, or that the rig has been validated on real data.
A contrastive encoder trained on structural metrics produced a non-collapsed latent space with statistically significant auto/manual separation (permutation $p < 0.001$, $N = 45$).	Any downstream accuracy, transfer, or generalisation claim.

Table 15: Explicit claim boundary. Claims in the left column are backed by governed evidence. Items in the right column are explicitly out of scope or unsupported by the collected evidence.

External validity. The study covers a single geographic area (Ingolstadt) using one OSM extract and one version of the pipeline. Generalization of the gap magnitudes (RMSE 54.84 m, Hausdorff 1,663 m) to other cities, OSM densities, or pipeline versions cannot be assumed. The manually authored reference map (Grid0828) is also a CARLA-format artifact produced by one team; differences in annotation conventions or map coverage policies could affect the gap magnitude independently of OSM quality. Additionally, the structural metrics quantify OpenDRIVE-level differences and do not account for differences at the simulation rendering level (mesh, textures, lighting), which are relevant for perception experiments.

Construct validity. The curvature KL divergence metric was found to approach zero despite a large spread difference, because both distributions are dominated by straight segments. This reflects a genuine construct limitation: KL divergence on marginal histograms is insensitive to distribution shape when the modes are aligned. The per-standard-deviation gap (0.865 vs. $0.055\,\mathrm{m}^{-1}$) is therefore the more informative construct for this comparison. Similarly, "CARLA load success" was used as a proxy for artifact quality until the planar ablation revealed it as insufficient. These construct issues are resolved in the reported analysis, but they illustrate why simulator load success is not a reliable quality proxy for OpenDRIVE artifacts.

Conclusion validity. The determinism conclusion is bounded: the 5-run audit shows topological invariance, but a larger audit across more versions or seed variations could reveal hidden nondeterminism. The perception conclusions are limited to infrastructure validation (Town10HD);

no direct effect size or statistical power calculation was performed for the deferred RQ3 experiments.

8.7. Extended GNN validation

Extended GNN validation ($N = 45$ tiles). To strengthen the observational GNN finding beyond the initial $N = 6$ pilot, a contrastive encoder was retrained on an expanded structural feature set of $N = 30$ auto-generated tiles (drawn from two pipeline runs across Ingolstadt extracts) and $N = 15$ manually authored tiles (drawn from Grid0828 runs). The encoder is a two-layer MLP trained with NT-Xent loss on nine structural features per tile (road count, junction count, mean and standard deviation of road length and curvature, lane count, connectivity rate, bounding-box area). The cross-cosine similarity between auto and manual embedding centroids is 0.124 (verdict: `NOT_COLLAPSED`). This is consistent with the primary graph-encoder result (cross-cosine $= -0.160$, $N = 72$ lane-graph tiles): in both cases the auto and manual tile embeddings occupy clearly separated regions of the latent space relative to the null distribution. A permutation test ($K = 1{,}000$ shuffles of the auto/manual labels) yields $p < 0.001$ (observed centroid cosine $= 0.124$ falls below all 1,000 permuted values; null distribution mean $= 0.877 \pm 0.067$). This confirms that the latent-space separation exceeds chance at $\alpha = 0.05$, upgrading RQ4 from purely observational to statistically supported.

8.8. Summary

The central lesson is that map conversion quality is not only a geometric-alignment problem. It is also a topology, junction, and importer-compatibility problem. The thesis therefore succeeds as a structural domain-gap study while explicitly deferring any claim that the generated Ingolstadt map is already a visually usable or perception-ready CARLA environment.

9. Conclusion

9.1. Main findings

This thesis presented a governed methodology for comparing automatically generated OSM→OpenDRIVE maps against a manually authored CARLA reference map of the same Ingolstadt region. The main contribution is an evidence-backed structural domain-gap workflow rather than a claim of full simulator readiness.

The authoritative run_11 comparison yields a whole-network RMSE of 54.84 m, a matched-subset RMSE of 24.47 m for 73.0% of sampled points within a 50 m criterion, and a Hausdorff distance of 1,663 m. The generated map is 4.88× longer in road coverage and contains 6.55× more junctions than the manual reference. Both multipliers reflect intentional scope differences rather than OSM-conversion errors: the auto-generated map encodes all OSM-tagged roads including residential streets, footways, and service roads that the manually-authored Grid0828 reference omitted; the junction ratio similarly reflects OSM's representation of every tagged intersection, including cul-de-sac and slip-road junctions absent from the manual baseline. Curvature spread is also substantially larger in the generated map (0.865 vs. 0.055 m^{-1}), indicating high-curvature outlier segments that are not present in the hand-authored map. These findings show that automatically generated city-scale maps can differ strongly from a manual reference even when both represent the same geographic area.

RQ1 is resolved in a bounded sense: the governed audit shows structural invariance in the measured topological counts under fixed inputs, while byte-level file hashes still diverge. RQ2 is resolved by the structural results above. RQ3 is bounded (rig confirmed on Town10HD; direct paired capture on the generated map deferred pending visual QA). RQ4 is statistically supported: a permutation test ($K = 1{,}000$) on the extended contrastive encoder ($N = 45$ tiles) yields $p < 0.001$, confirming that the latent-space separation between auto-generated and manually authored tiles exceeds chance; no transfer-feasibility claim is made. RQ5 remains deferred (Munich second-city probe completed; full paired transfer experiment not executed).

The practical implication is that OSM-derived CARLA generation can support scalable structural experimentation, but the present evidence does not justify replacing manually curated maps for perception-ready validation without additional topology repair, visual QA, and paired sensor-domain evaluation. This result directly informs the make-or-automate decision: automated generation is sufficient for early structural screening and reproducibility studies, while perception experiments and training-data generation require a manually curated or post-repaired map that has passed visual QA and been packaged with semantic material tags.

9.2. Final thesis boundary

The final governed evidence also supports an important negative result. The generated XODR can be parsed and loaded by CARLA, but visual QA of the road mesh fails. A controlled planar ablation showed that the original DEM-backed artifact and both flattened variants continue to exhibit road-mesh and surface-continuity defects, so the dominant defect is not primarily caused by elevation. This means that CARLA load success cannot be treated as proof of visual usability or perception readiness.

Accordingly, the thesis does not claim positive generated-map visual evidence or generated-map perception evidence. Town10HD remains valid only as a bounded demonstration that the governed sensor-rig and capture mechanics work on a stable built-in map. The structural comparison remains valid because it is performed at the OpenDRIVE level and does not depend on visually successful CARLA rendering.

9.3. Future work

The evidence boundary established by this thesis motivates the following concrete directions.

1. Junction topology repair. The most impactful near-term step is a dedicated junction-reconstruction pass for complex OSM-derived junction regions, in particular roundabouts. Roundabouts require more than geometric smoothing: entry and exit connectors, circular carriageway representation, lane-link consistency, priority/yield semantics, and CARLA importer compatibility must be generated and validated as a coherent topology. Such work should be evaluated in a separate governed rerun with explicit artifact provenance and should remain distinct from the authoritative run_11 structural results reported in this thesis.

2. Direct perceptual domain-gap measurement (RQ3). Once the generated map passes visual QA, a paired perception experiment should be executed: the same sensor rig and route protocol used for Town10HD should be applied to the generated Ingolstadt map and to the manually modeled Grid0821/Grid0828 reference, and distribution-level proxy metrics (e.g., pixel-class histogram shift, depth-distribution KL divergence) should be computed. This experiment was the original intent of RQ3 and represents the most direct continuation of the thesis work.

3. Multi-city structural gap characterisation. The current study is limited to a single city and OSM extract. Replicating the structural gap analysis on two or three additional cities with different OSM density profiles (e.g., a rural area and a dense urban center outside Germany) would allow the RMSE and Hausdorff gap magnitudes to be interpreted as characteristic of OSM map quality in general versus specific to Ingolstadt.

4. DEM-backed elevation as a first-class evaluation dimension. The planar ablation in this thesis confirms that DEM elevation is not the dominant explanation for the current CARLA visual failure, but the primary structural analysis was also restricted to a flat artifact. After the junction topology layer stabilises, DEM-backed maps should be evaluated in a matched flat-versus-DEM comparison to quantify the elevation-specific contribution to both visual QA and structural metrics.

5. GNN latent-space transfer experiment (RQ4/RQ5). The observational GNN result (cross-cosine $= -0.160$, non-collapsed) suggests that the structural metrics contain domain-discriminative information. A natural continuation is to train a downstream classifier or retrieval model on the latent embeddings and evaluate transfer to an unseen city, providing a more principled answer to RQ4 and a first step toward RQ5.

6. Semantic enrichment and object-class gap closure. The four-category semantic-object gap (zero buildings, barriers, poles, trees in the contract-run artifact) was partially closed by a later pipeline fix. A systematic evaluation of semantic completeness — including signal density, lamp-post placement, and building footprint accuracy — would complement the geometric gap analysis and is relevant for scene-diversity studies in synthetic training data generation.

7. Automated map-readiness gate integration. The governed pipeline contains a layered visual-readiness audit stack—connector-pose mismatch detection (`audit_xodr_visual_geometry`), CARLA visual smoke testing (`carla_visual_smoke_gate`), and a combined readiness gate (`final_map_readiness_gate`)—developed in response to the finding that CARLA load success is a necessary but insufficient condition for visual and perceptual usability. A natural engineering direction is to promote this gate stack to a mandatory governed checkpoint: no perception experiment should be allowed to start unless the map has passed connector-gap thresholds, visual road-mesh QA, and spawn-point count verification. This would close the current gap between `CARLA_LOADABLE = yes` and `CARLA_VISUAL_READY = yes` in a repeatable, artifact-logged way, and would make the readiness classification derivable directly from the pipeline output rather than from post-hoc manual inspection.

8. Manually authored second-city reference for RQ5. The Munich OSM probe shows that the map-generation path can process a second city, but it does not complete RQ5 because no manually authored Munich OpenDRIVE reference exists. A manually authored Munich reference, built over the same spatial extent as the OSM extract, would enable a true paired cross-city structural comparison and would be the cleanest way to close the generalisation question.

9. Full semantic segmentation through UE4 map cook. Generated-map semantic segmentation remains bounded by the CARLA standalone OpenDRIVE material-tag limitation. Resolving this requires importing the generated map into the Unreal Engine editor workflow, assigning semantic material tags, cooking it as a full CARLA package, and then rerunning the identical sensor rig. This is substantially larger than an OpenDRIVE-only repair pass and should be treated as a separate future-work track.

10. Non-rigid structural alignment. The authoritative run_11 alignment is rigid SE(2) with fixed scale. This is appropriate for a governed baseline, but it cannot account for local distortions caused by road decomposition, OSM polyline simplification, or manual-map generalisation. A future comparison could evaluate non-rigid warping, local ICP, or mesh-level correspondence methods to separate global misalignment from local geometric deformation.

11. RQ4 scale-up. The post-submission structural-feature separation result is a small-N pilot. A statistically meaningful RQ4 experiment should scale from $N = 6$ to at least $N \geq 30$ samples per class, spanning multiple cities, multiple generated-map versions, and multiple manual references where available. Only then should classifier accuracy, silhouette scores, and latent-space separability be interpreted as robust rather than observational.

12. Traffic-light inference heuristic. Traffic-light and signal enrichment are not established thesis results because final inspected artifacts contain zero verified functional signal insertions. Future work should tune or replace the current traffic-light inference heuristic, which appears too conservative for the available OSM-derived junction patterns, and should require artifact-level evidence of inserted `<signal>` and `<signalReference>` elements before claiming traffic-control enrichment.

13. Byte-level determinism root cause. The 5-run determinism audit shows topological stability ($CV = 0.0$ for road and junction counts) but byte-level XODR hash divergence. Future work should isolate whether the remaining divergence comes from floating-point accumulation, nondeterministic sort order, timestamps, UUIDs, XML serialisation order, or external tool behaviour. This would upgrade the RQ1 result from bounded structural determinism toward byte-identical reproducibility.

14. Fréchet and shape-sensitive distance metrics. RMSE and Hausdorff distance are useful governed baselines, but they are sensitive to sampling and road-decomposition mismatch. A curve-aware metric such as discrete Fréchet distance, computed after robust road-segment correspondence, could provide a more shape-sensitive comparison of manual and generated road centerlines. The main open challenge is defining correspondence when the auto and manual maps decompose the same road corridor into different numbers of OpenDRIVE roads.

15. Realism augmentation and perception-quality validation. The pipeline contains realism and enrichment infrastructure for visual detail, object placement, and potential domain randomisation, but this thesis does not validate those modules as perception-quality improvements. Future work should evaluate realism augmentation only after the generated map passes visual QA and is packaged as a full CARLA asset with valid semantic material tags. A defensible study would compare identical sensor routes with and without realism augmentation, quantify changes in semantic segmentation, RGB, depth, and object-distribution statistics, and test whether the added realism reduces the measured perceptual domain gap rather than merely increasing scene complexity.

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A. Reproducibility protocol and run evidence

A.1. Canonical Geographic Bounds (Ingolstadt)

All experiments are constrained to the following fixed OSM bounding box:

- Lat-Min: 48.74935649548228
- Lon-Min: 11.422268084715878
- Lat-Max: 48.77444431571603
- Lon-Max: 11.47882091528412

A.2. Authoritative Sensor Rig Contract

To ensure mathematically consistent data across domains, the following rig contract is enforced:

- **Camera Intrinsics:** Use `K_undistortion` for an ideal pinhole model. Original K and D (distortion) are ignored.
- **Camera Extrinsics:** The vehicle-to-camera transform (cTv) is used directly (not inverted).
- **LiDAR Extrinsics:** The LiDAR-to-vehicle transform (vTl) MUST be inverted for attachment in CARLA's frame.

A.3. Stable Evidence Roots

The quantitative findings in this thesis are backed exclusively by artifacts in the following stable directories:

- `artifacts/final_runs/scenario_b_audit/evidence/` (Audited thesis evidence)
- `thesis_results/` (Aggregated structural gap and variability reports)

A.4. Minimum run manifest

A complete audit log of the governed pipeline stages, artifact hashes, and environment settings is provided in the `thesis_results/` directory under each run's manifest JSON.

B. Additional Reference Material

B.1. Pipeline entry points

The full pipeline is implemented as a staged toolchain with explicit entry points for reproducible execution and auditing. The following scripts are the intended "front doors" of the code base:

- `run_pipeline.py`: executes the end-to-end OSM→OpenDRIVE→CARLA pipeline and writes a self-contained run directory.
- `main_pipeline.py`: pipeline orchestrator implementing the ordered stages and their artifact contracts.
- `run_quality_gates.py`: runs the quality-gate suite on an existing run folder, producing structured validation reports and a failure taxonomy.
- `run_determinism_audit.py`: external two-run audit that repeats the pipeline and compares hashed artifacts to determine whether the pipeline is deterministic under fixed inputs and settings.
- `run_full_domain_gap.py`: performs offline domain-gap evaluation between an automatically generated map and a manually authored reference map, including alignment, tile matching, and structural/perceptual gap summaries.

B.2. Sensor calibration contract

All perception experiments use a fixed, ego-mounted sensor rig defined in `calib_data.json`. Camera intrinsics are taken from `K_undistortion` and treated as a pinhole model. The distortion parameters `K` and `D` are ignored, and no additional undistortion step is applied to rendered images. Extrinsics are interpreted using the provided frame conventions:

- `cTv`: vehicle→camera transform (used to place camera sensors consistently).
- `vTl`: LiDAR→vehicle transform (inverted if the simulator API expects vehicle→LiDAR).

These conventions prevent calibration errors from confounding the domain-gap analysis.

B.3. Run artifact inventory

Each pipeline run produces a directory containing stage-wise outputs and diagnostic artifacts. Table 16 lists the most important artifacts referenced throughout the thesis.

Table 16: Key artifacts produced per run directory.

Artifact	Purpose
`settings_snapshot.json` + hash	frozen configuration and toolchain identity
`final.xodr` / tile `*.xodr`	authoritative OpenDRIVE map(s)
`map_statistics.json`	descriptive statistics (roads, lanes, junctions, semantics)
`validation_report_full.json`	structured gate results and invariant violations
`gate_failures.json`	concise list of failed gates for fast regression checks
`xodr_negative_s_report.csv`	longitudinal-s invariant violations and repairs
`tile_metadata.json`, `tile_adjacency.json`	deterministic tiling layout and adjacency graph
`tile_qa_reports/*.json`	per-tile CARLA import/spawn/drive checks
`tile_failure_taxonomy.json`	aggregated failure categories (map vs simulator)
`domain_gap/full_report.json`	offline gap metrics and aligned overlays
`figures/*`	diagnostic images (overlays, heatmaps, continuity plots)

B.4. Submission checklist (THI formal requirements)

Final administrative metadata must be verified against the official thesis registration record before submission.